**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

April 16, 2024

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Biosolids
Pace Project No.: 50369587

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on April 02, 2024. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures

**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# CERTIFICATIONS

| | |
|---|---|
| Project: | Biosolids |
| Pace Project No.: | 50369587 |

**Pace Analytical Services Indianapolis**

7726 Moller Road, Indianapolis, IN 46268
Illinois Accreditation #: 200074
Indiana Drinking Water Laboratory #: C-49-06
Kansas/TNI Certification #: E-10177
Kentucky UST Agency Interest #: 80226
Kentucky WW Laboratory ID #: 98019
Michigan Drinking Water Laboratory #9050
Ohio VAP Certified Laboratory #: CL0065
Oklahoma Laboratory #: 9204
Texas Certification #: T104704355
Washington Dept of Ecology #: C1081
Wisconsin Laboratory #: 999788130
USDA Foreign Soil Permit #: 525-23-13-23119
USDA Compliance Agreement #: IN-SL-22-001

**Pace Analytical Services, LLC**
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Grand Rapids, MI 49512
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# SAMPLE SUMMARY

Project: Biosolids
Pace Project No.: 50369587

| Lab ID | Sample ID | Matrix | Date Collected | Date Received |
|---|---|---|---|---|
| **50369587001** | **DIGESTOR #4** | Solid | 04/01/24 14:15 | 04/02/24 10:04 |

**Pace Analytical Services, LLC**
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Grand Rapids, MI 49512
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## SAMPLE ANALYTE COUNT

Project: Biosolids
Pace Project No.: 50369587

| Lab ID | Sample ID | Method | Analysts | Analytes Reported | Laboratory |
|---|---|---|---|---|---|
| **50369587001** | **DIGESTOR #4** | EPA 8081 | KAV | 21 | PASI-I |
| | | EPA 8081 | KAV | 8 | PASI-I |
| | | EPA 8082 | AM | 8 | PASI-I |
| | | EPA 8151A | CPH | 3 | PASI-I |
| | | EPA 6010 | ELK | 6 | PASI-I |
| | | EPA 6010 | ELK | 8 | PASI-I |
| | | EPA 6020 | MTM | 10 | PASI-I |
| | | EPA 7470 | ILP | 1 | PASI-I |
| | | EPA 7471 | EAE | 1 | PASI-I |
| | | EPA 8270 | JCM | 67 | PASI-I |
| | | EPA 8270 | FIP | 18 | PASI-I |
| | | EPA 8260 | TMW | 55 | PASI-I |
| | | EPA 5030/8260 | BES | 13 | PASI-I |
| | | SM 2540G | QAK | 1 | PASI-I |
| | | SM 2540G | QAK | 1 | PASI-I |
| | | SM 2540G | QAK | 1 | PASI-I |
| | | EPA 9038 | BEP | 1 | PASI-I |
| | | EPA 9045 | BEP | 1 | PASI-I |
| | | EPA 351.2 | OAS | 1 | PASI-I |
| | | EPA 353.2 | DAW | 2 | PASI-I |
| | | SM 4500-NH3 G | OAS | 1 | PASI-I |
| | | SM 4500-Cl-E | ZM | 1 | PASI-I |
| | | SM 4500-CN-E | ZM | 1 | PASI-I |

PASI-I = Pace Analytical Services - Indianapolis

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# ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50369587

**Sample: DIGESTOR #4** **Lab ID: 50369587001** Collected: 04/01/24 14:15 Received: 04/02/24 10:04 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **8081 GCS Pesticide Solids** | Analytical Method: EPA 8081 Preparation Method: EPA 3546 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| Aldrin | **<983** | ug/kg | 983 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 309-00-2 | |
| alpha-BHC | **<983** | ug/kg | 983 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 319-84-6 | |
| beta-BHC | **<983** | ug/kg | 983 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 319-85-7 | |
| delta-BHC | **<983** | ug/kg | 983 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 319-86-8 | |
| gamma-BHC (Lindane) | **<983** | ug/kg | 983 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 58-89-9 | |
| Chlordane (Technical) | **<19700** | ug/kg | 19700 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 57-74-9 | |
| alpha-Chlordane | **<983** | ug/kg | 983 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 5103-71-9 | |
| gamma-Chlordane | **<983** | ug/kg | 983 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 5103-74-2 | |
| 4,4'-DDD | **<1970** | ug/kg | 1970 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 72-54-8 | |
| 4,4'-DDE | **<1970** | ug/kg | 1970 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 72-55-9 | |
| 4,4'-DDT | **<1970** | ug/kg | 1970 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 50-29-3 | |
| Dieldrin | **<1970** | ug/kg | 1970 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 60-57-1 | |
| Endosulfan I | **<983** | ug/kg | 983 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 959-98-8 | |
| Endosulfan II | **<1970** | ug/kg | 1970 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 33213-65-9 | |
| Endosulfan sulfate | **<1970** | ug/kg | 1970 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 1031-07-8 | |
| Endrin | **<1970** | ug/kg | 1970 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 72-20-8 | |
| Endrin aldehyde | **<1970** | ug/kg | 1970 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 7421-93-4 | |
| Heptachlor | **<983** | ug/kg | 983 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 76-44-8 | |
| Heptachlor epoxide | **<983** | ug/kg | 983 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 1024-57-3 | |
| Toxaphene | **<19700** | ug/kg | 19700 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 8001-35-2 | |
| ***Surrogates*** | | | | | | | | |
| Decachlorobiphenyl (S) | 36 | %. | 10-166 | 1 | 04/04/24 12:18 | 04/05/24 01:04 | 2051-24-3 | |
| **8081 GCS Pest RV, TCLP** | Analytical Method: EPA 8081 Preparation Method: EPA 3510 | | | | | | | |
| | Leachate Method/Date: EPA 1311; 04/07/24 22:45 Initial pH: 8.36; Final pH: 5.13 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| gamma-BHC (Lindane) | **<0.00025** | mg/L | 0.00025 | 1 | 04/09/24 16:52 | 04/10/24 11:58 | 58-89-9 | |
| Chlordane (Technical) | **<0.0050** | mg/L | 0.0050 | 1 | 04/09/24 16:52 | 04/10/24 11:58 | 57-74-9 | |
| Endrin | **<0.00050** | mg/L | 0.00050 | 1 | 04/09/24 16:52 | 04/10/24 11:58 | 72-20-8 | |
| Heptachlor | **<0.00025** | mg/L | 0.00025 | 1 | 04/09/24 16:52 | 04/10/24 11:58 | 76-44-8 | |
| Heptachlor epoxide | **<0.00025** | mg/L | 0.00025 | 1 | 04/09/24 16:52 | 04/10/24 11:58 | 1024-57-3 | |
| Methoxychlor | **<0.0025** | mg/L | 0.0025 | 1 | 04/09/24 16:52 | 04/10/24 11:58 | 72-43-5 | |
| Toxaphene | **<0.0050** | mg/L | 0.0050 | 1 | 04/09/24 16:52 | 04/10/24 11:58 | 8001-35-2 | |
| ***Surrogates*** | | | | | | | | |
| Decachlorobiphenyl (S) | 35 | %. | 10-124 | 1 | 04/09/24 16:52 | 04/10/24 11:58 | 2051-24-3 | |
| **8082 PCB Solids** | Analytical Method: EPA 8082 Preparation Method: EPA 3546 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| PCB-1016 (Aroclor 1016) | **<29500** | ug/kg | 29500 | 1 | 04/12/24 11:07 | 04/13/24 04:02 | 12674-11-2 | |
| PCB-1221 (Aroclor 1221) | **<29500** | ug/kg | 29500 | 1 | 04/12/24 11:07 | 04/13/24 04:02 | 11104-28-2 | |
| PCB-1232 (Aroclor 1232) | **<29500** | ug/kg | 29500 | 1 | 04/12/24 11:07 | 04/13/24 04:02 | 11141-16-5 | |
| PCB-1242 (Aroclor 1242) | **<29500** | ug/kg | 29500 | 1 | 04/12/24 11:07 | 04/13/24 04:02 | 53469-21-9 | |
| PCB-1248 (Aroclor 1248) | **<29500** | ug/kg | 29500 | 1 | 04/12/24 11:07 | 04/13/24 04:02 | 12672-29-6 | |
| PCB-1254 (Aroclor 1254) | **<29500** | ug/kg | 29500 | 1 | 04/12/24 11:07 | 04/13/24 04:02 | 11097-69-1 | |
| PCB-1260 (Aroclor 1260) | **<29500** | ug/kg | 29500 | 1 | 04/12/24 11:07 | 04/13/24 04:02 | 11096-82-5 | |

## REPORT OF LABORATORY ANALYSIS

**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50369587

**Sample: DIGESTOR #4** **Lab ID: 50369587001** Collected: 04/01/24 14:15 Received: 04/02/24 10:04 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **8082 PCB Solids** | Analytical Method: EPA 8082 Preparation Method: EPA 3546 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| ***Surrogates*** | | | | | | | | |
| Tetrachloro-m-xylene (S) | 68 | %. | 10-133 | 1 | 04/12/24 11:07 | 04/13/24 04:02 | 877-09-8 | |
| **8151A Cl Acid Herbicides TCLP** | Analytical Method: EPA 8151A Preparation Method: EPA 8151 | | | | | | | |
| | Leachate Method/Date: EPA 1311; 04/07/24 22:45 Initial pH: 8.36; Final pH: 5.13 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| 2,4-D | **<0.0050** | mg/L | 0.0050 | 1 | 04/11/24 21:27 | 04/13/24 18:20 | 94-75-7 | |
| 2,4,5-TP (Silvex) | **<0.0050** | mg/L | 0.0050 | 1 | 04/11/24 21:27 | 04/13/24 18:20 | 93-72-1 | |
| ***Surrogates*** | | | | | | | | |
| 2,4-DCAA (S) | 65 | %. | 22-132 | 1 | 04/11/24 21:27 | 04/13/24 18:20 | 19719-28-9 | |
| **6010 MET ICP** | Analytical Method: EPA 6010 Preparation Method: EPA 3050 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| Aluminum | **3930** | mg/kg | 2600 | 1 | 04/09/24 21:14 | 04/10/24 12:33 | 7429-90-5 | |
| Calcium | **33900** | mg/kg | 2600 | 1 | 04/09/24 21:14 | 04/10/24 12:33 | 7440-70-2 | |
| Magnesium | **6340** | mg/kg | 2600 | 1 | 04/09/24 21:14 | 04/10/24 12:33 | 7439-95-4 | |
| Phosphorus | **37000** | mg/kg | 2600 | 1 | 04/09/24 21:14 | 04/10/24 12:33 | 7723-14-0 | N2 |
| Potassium | **5740** | mg/kg | 2600 | 1 | 04/09/24 21:14 | 04/10/24 12:33 | 7440-09-7 | |
| Sodium | **6670** | mg/kg | 2600 | 1 | 04/09/24 21:14 | 04/10/24 12:33 | 7440-23-5 | |
| **6010 MET ICP, TCLP** | Analytical Method: EPA 6010 Preparation Method: EPA 3010 | | | | | | | |
| | Leachate Method/Date: EPA 1311; 04/07/24 22:45 Initial pH: 8.36; Final pH: 5.13 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| Arsenic | **<0.10** | mg/L | 0.10 | 1 | 04/09/24 21:07 | 04/10/24 09:57 | 7440-38-2 | |
| Barium | **<5.0** | mg/L | 5.0 | 1 | 04/09/24 21:07 | 04/10/24 09:57 | 7440-39-3 | |
| Cadmium | **<0.050** | mg/L | 0.050 | 1 | 04/09/24 21:07 | 04/10/24 09:57 | 7440-43-9 | |
| Chromium | **<0.10** | mg/L | 0.10 | 1 | 04/09/24 21:07 | 04/10/24 09:57 | 7440-47-3 | |
| Lead | **<0.10** | mg/L | 0.10 | 1 | 04/09/24 21:07 | 04/10/24 09:57 | 7439-92-1 | |
| Selenium | **<0.10** | mg/L | 0.10 | 1 | 04/09/24 21:07 | 04/10/24 09:57 | 7782-49-2 | |
| Silver | **<0.10** | mg/L | 0.10 | 1 | 04/09/24 21:07 | 04/10/24 09:57 | 7440-22-4 | |
| Zinc | **3.9** | mg/L | 0.20 | 1 | 04/09/24 21:07 | 04/10/24 09:57 | 7440-66-6 | |
| **6020 MET ICPMS** | Analytical Method: EPA 6020 Preparation Method: EPA 3050B | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| Arsenic | **18.2** | mg/kg | 6.0 | 1 | 04/10/24 15:42 | 04/12/24 04:20 | 7440-38-2 | |
| Cadmium | **7.3** | mg/kg | 3.0 | 1 | 04/10/24 15:42 | 04/12/24 04:20 | 7440-43-9 | |
| Chromium | **123** | mg/kg | 12.1 | 1 | 04/10/24 15:42 | 04/12/24 04:20 | 7440-47-3 | |
| Copper | **818** | mg/kg | 6.0 | 1 | 04/10/24 15:42 | 04/12/24 04:20 | 7440-50-8 | |
| Lead | **16.6** | mg/kg | 6.0 | 1 | 04/10/24 15:42 | 04/12/24 04:20 | 7439-92-1 | |
| Molybdenum | **21.0** | mg/kg | 6.0 | 1 | 04/10/24 15:42 | 04/12/24 04:20 | 7439-98-7 | N2 |
| Nickel | **26.5** | mg/kg | 6.0 | 1 | 04/10/24 15:42 | 04/12/24 04:20 | 7440-02-0 | |
| Selenium | **<6.0** | mg/kg | 6.0 | 1 | 04/10/24 15:42 | 04/12/24 04:20 | 7782-49-2 | |
| Silver | **4.7** | mg/kg | 3.0 | 1 | 04/10/24 15:42 | 04/12/24 04:20 | 7440-22-4 | |
| Zinc | **2420** | mg/kg | 151 | 5 | 04/10/24 15:42 | 04/12/24 03:56 | 7440-66-6 | |

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# ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50369587

**Sample: DIGESTOR #4** **Lab ID: 50369587001** Collected: 04/01/24 14:15 Received: 04/02/24 10:04 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **7470 Mercury, TCLP** | Analytical Method: EPA 7470 Preparation Method: EPA 7470<br>Leachate Method/Date: EPA 1311; 04/07/24 22:45 Initial pH: 8.36; Final pH: 5.13<br>Pace Analytical Services - Indianapolis | | | | | | | |
| Mercury | **<0.0020** | mg/L | 0.0020 | 1 | 04/10/24 19:48 | 04/11/24 09:59 | 7439-97-6 | |
| **7471 Mercury** | Analytical Method: EPA 7471 Preparation Method: EPA 7471<br>Pace Analytical Services - Indianapolis | | | | | | | |
| Mercury | **<12.8** | mg/kg | 12.8 | 1 | 04/11/24 12:00 | 04/11/24 19:22 | 7439-97-6 | |
| **8270 SVOC SS Soil** | Analytical Method: EPA 8270 Preparation Method: EPA 3546<br>Pace Analytical Services - Indianapolis | | | | | | | |
| Acenaphthene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 83-32-9 | |
| Acenaphthylene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 208-96-8 | |
| Anthracene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 120-12-7 | |
| Benzo(a)anthracene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 56-55-3 | |
| Benzo(a)pyrene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 50-32-8 | |
| Benzo(b)fluoranthene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 205-99-2 | |
| Benzo(g,h,i)perylene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 191-24-2 | |
| Benzo(k)fluoranthene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 207-08-9 | |
| Benzyl alcohol | **<389000** | ug/kg | 389000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 100-51-6 | |
| 4-Bromophenylphenyl ether | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 101-55-3 | |
| Butylbenzylphthalate | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 85-68-7 | |
| 4-Chloro-3-methylphenol | **<389000** | ug/kg | 389000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 59-50-7 | |
| 4-Chloroaniline | **<389000** | ug/kg | 389000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 106-47-8 | |
| bis(2-Chloroethoxy)methane | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 111-91-1 | |
| bis(2-Chloroethyl) ether | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 111-44-4 | |
| bis(2chloro1methylethyl) ether | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 108-60-1 | |
| 2-Chloronaphthalene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 91-58-7 | |
| 2-Chlorophenol | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 95-57-8 | |
| 4-Chlorophenylphenyl ether | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 7005-72-3 | |
| Chrysene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 218-01-9 | |
| Dibenz(a,h)anthracene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 53-70-3 | |
| Dibenzofuran | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 132-64-9 | |
| 3,3'-Dichlorobenzidine | **<389000** | ug/kg | 389000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 91-94-1 | |
| 2,4-Dichlorophenol | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 120-83-2 | |
| Diethylphthalate | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 84-66-2 | |
| 2,4-Dimethylphenol | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 105-67-9 | |
| Dimethylphthalate | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 131-11-3 | |
| Di-n-butylphthalate | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 84-74-2 | |
| 4,6-Dinitro-2-methylphenol | **<389000** | ug/kg | 389000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 534-52-1 | |
| 2,4-Dinitrophenol | **<944000** | ug/kg | 944000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 51-28-5 | |
| 2,4-Dinitrotoluene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 121-14-2 | |
| 2,6-Dinitrotoluene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 606-20-2 | |
| Di-n-octylphthalate | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 117-84-0 | |
| bis(2-Ethylhexyl)phthalate | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 117-81-7 | |
| Fluoranthene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 206-44-0 | |

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# ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50369587

**Sample: DIGESTOR #4** **Lab ID: 50369587001** Collected: 04/01/24 14:15 Received: 04/02/24 10:04 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **8270 SVOC SS Soil** | Analytical Method: EPA 8270 Preparation Method: EPA 3546 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| Fluorene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 86-73-7 | |
| Hexachloro-1,3-butadiene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 87-68-3 | |
| Hexachlorobenzene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 118-74-1 | |
| Hexachlorocyclopentadiene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 77-47-4 | |
| Hexachloroethane | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 67-72-1 | |
| Indeno(1,2,3-cd)pyrene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 193-39-5 | |
| Isophorone | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 78-59-1 | |
| 1-Methylnaphthalene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 90-12-0 | |
| 2-Methylnaphthalene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 91-57-6 | |
| 2-Methylphenol(o-Cresol) | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 95-48-7 | |
| 3&4-Methylphenol(m&p Cresol) | **<389000** | ug/kg | 389000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | | |
| Naphthalene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 91-20-3 | |
| 2-Nitroaniline | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 88-74-4 | |
| 3-Nitroaniline | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 99-09-2 | |
| 4-Nitroaniline | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 100-01-6 | |
| Nitrobenzene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 98-95-3 | |
| 2-Nitrophenol | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 88-75-5 | |
| 4-Nitrophenol | **<944000** | ug/kg | 944000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 100-02-7 | |
| N-Nitroso-di-n-propylamine | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 621-64-7 | |
| N-Nitrosodiphenylamine | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 86-30-6 | |
| Pentachlorophenol | **<944000** | ug/kg | 944000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 87-86-5 | |
| Phenanthrene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 85-01-8 | |
| Phenol | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 108-95-2 | P1 |
| Pyrene | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 129-00-0 | |
| 2,4,5-Trichlorophenol | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 95-95-4 | |
| 2,4,6-Trichlorophenol | **<195000** | ug/kg | 195000 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 88-06-2 | |
| ***Surrogates*** | | | | | | | | |
| Nitrobenzene-d5 (S) | 80 | %. | 32-95 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 4165-60-0 | |
| Phenol-d5 (S) | 87 | %. | 27-116 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 4165-62-2 | |
| 2-Fluorophenol (S) | 80 | %. | 21-109 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 367-12-4 | |
| 2,4,6-Tribromophenol (S) | 77 | %. | 10-121 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 118-79-6 | |
| 2-Fluorobiphenyl (S) | 68 | %. | 33-102 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 321-60-8 | |
| p-Terphenyl-d14 (S) | 87 | %. | 20-120 | 1 | 04/08/24 14:24 | 04/11/24 16:52 | 1718-51-0 | |
| **8270 MSSV TCLP Sep Funnel** | Analytical Method: EPA 8270 Preparation Method: EPA 3510 | | | | | | | |
| | Leachate Method/Date: EPA 1311; 04/07/24 22:45 Initial pH: 8.36; Final pH: 5.13 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| 1,4-Dichlorobenzene | **<0.10** | mg/L | 0.10 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 106-46-7 | |
| 2,4-Dinitrotoluene | **<0.10** | mg/L | 0.10 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 121-14-2 | |
| Hexachloro-1,3-butadiene | **<0.10** | mg/L | 0.10 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 87-68-3 | |
| Hexachlorobenzene | **<0.10** | mg/L | 0.10 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 118-74-1 | |
| Hexachloroethane | **<0.10** | mg/L | 0.10 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 67-72-1 | |
| 2-Methylphenol(o-Cresol) | **<0.10** | mg/L | 0.10 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 95-48-7 | |
| 3&4-Methylphenol(m&p Cresol) | **<0.20** | mg/L | 0.20 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | | |
| Nitrobenzene | **<0.10** | mg/L | 0.10 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 98-95-3 | |

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# ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50369587

**Sample: DIGESTOR #4** **Lab ID: 50369587001** Collected: 04/01/24 14:15 Received: 04/02/24 10:04 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **8270 MSSV TCLP Sep Funnel** | Analytical Method: EPA 8270 Preparation Method: EPA 3510 | | | | | | | |
| | Leachate Method/Date: EPA 1311; 04/07/24 22:45 Initial pH: 8.36; Final pH: 5.13 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| Pentachlorophenol | **<0.50** | mg/L | 0.50 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 87-86-5 | |
| Pyridine | **<0.10** | mg/L | 0.10 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 110-86-1 | |
| 2,4,5-Trichlorophenol | **<0.50** | mg/L | 0.50 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 95-95-4 | |
| 2,4,6-Trichlorophenol | **<0.10** | mg/L | 0.10 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 88-06-2 | |
| ***Surrogates*** | | | | | | | | |
| Nitrobenzene-d5 (S) | 80 | %. | 44-107 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 4165-60-0 | |
| 2-Fluorobiphenyl (S) | 60 | %. | 36-92 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 321-60-8 | |
| p-Terphenyl-d14 (S) | 87 | %. | 57-146 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 1718-51-0 | |
| Phenol-d5 (S) | 35 | %. | 15-47 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 4165-62-2 | |
| 2-Fluorophenol (S) | 47 | %. | 23-69 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 367-12-4 | |
| 2,4,6-Tribromophenol (S) | 88 | %. | 37-127 | 1 | 04/10/24 12:57 | 04/12/24 11:54 | 118-79-6 | |
| **8260 MSV 5030 Low Level** | Analytical Method: EPA 8260 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| Acetone | **<6.1** | mg/kg | 6.1 | 1 | | 04/05/24 17:43 | 67-64-1 | |
| Acrolein | **<6.1** | mg/kg | 6.1 | 1 | | 04/05/24 17:43 | 107-02-8 | |
| Acrylonitrile | **<6.1** | mg/kg | 6.1 | 1 | | 04/05/24 17:43 | 107-13-1 | |
| Benzene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 71-43-2 | |
| Bromodichloromethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 75-27-4 | |
| Bromoform | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 75-25-2 | |
| Bromomethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 74-83-9 | L1 |
| 2-Butanone (MEK) | **<1.5** | mg/kg | 1.5 | 1 | | 04/05/24 17:43 | 78-93-3 | |
| Carbon disulfide | **<0.61** | mg/kg | 0.61 | 1 | | 04/05/24 17:43 | 75-15-0 | |
| Carbon tetrachloride | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 56-23-5 | |
| Chlorobenzene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 108-90-7 | |
| Chloroethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 75-00-3 | |
| Chloroform | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 67-66-3 | |
| Chloromethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 74-87-3 | |
| 1,2-Dibromo-3-chloropropane | **<0.61** | mg/kg | 0.61 | 1 | | 04/05/24 17:43 | 96-12-8 | |
| Dibromochloromethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 124-48-1 | |
| 1,2-Dibromoethane (EDB) | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 106-93-4 | |
| Dibromomethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 74-95-3 | |
| 1,2-Dichlorobenzene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 95-50-1 | |
| 1,3-Dichlorobenzene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 541-73-1 | |
| 1,4-Dichlorobenzene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 106-46-7 | |
| trans-1,4-Dichloro-2-butene | **<6.1** | mg/kg | 6.1 | 1 | | 04/05/24 17:43 | 110-57-6 | |
| Dichlorodifluoromethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 75-71-8 | |
| 1,1-Dichloroethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 75-34-3 | |
| 1,2-Dichloroethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 107-06-2 | |
| 1,1-Dichloroethene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 75-35-4 | |
| cis-1,2-Dichloroethene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 156-59-2 | |
| trans-1,2-Dichloroethene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 156-60-5 | |
| 1,2-Dichloropropane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 78-87-5 | |
| cis-1,3-Dichloropropene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 10061-01-5 | |

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# ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50369587

**Sample: DIGESTOR #4** **Lab ID: 50369587001** Collected: 04/01/24 14:15 Received: 04/02/24 10:04 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **8260 MSV 5030 Low Level** | Analytical Method: EPA 8260 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| trans-1,3-Dichloropropene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 10061-02-6 | |
| Ethylbenzene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 100-41-4 | |
| Ethyl methacrylate | **<6.1** | mg/kg | 6.1 | 1 | | 04/05/24 17:43 | 97-63-2 | |
| 2-Hexanone | **<6.1** | mg/kg | 6.1 | 1 | | 04/05/24 17:43 | 591-78-6 | |
| Iodomethane | **<6.1** | mg/kg | 6.1 | 1 | | 04/05/24 17:43 | 74-88-4 | |
| Methylene Chloride | **<1.2** | mg/kg | 1.2 | 1 | | 04/05/24 17:43 | 75-09-2 | |
| Methyl methacrylate | **<6.1** | mg/kg | 6.1 | 1 | | 04/05/24 17:43 | 80-62-6 | |
| 4-Methyl-2-pentanone (MIBK) | **<1.5** | mg/kg | 1.5 | 1 | | 04/05/24 17:43 | 108-10-1 | |
| Methyl-tert-butyl ether | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 1634-04-4 | |
| Styrene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 100-42-5 | |
| 1,1,1,2-Tetrachloroethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 630-20-6 | |
| 1,1,2,2-Tetrachloroethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 79-34-5 | |
| Tetrachloroethene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 127-18-4 | |
| Toluene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 108-88-3 | |
| 1,1,1-Trichloroethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 71-55-6 | |
| 1,1,2-Trichloroethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 79-00-5 | |
| Trichloroethene | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 79-01-6 | |
| Trichlorofluoromethane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 75-69-4 | |
| 1,2,3-Trichloropropane | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 96-18-4 | |
| Vinyl acetate | **<6.1** | mg/kg | 6.1 | 1 | | 04/05/24 17:43 | 108-05-4 | |
| Vinyl chloride | **<0.30** | mg/kg | 0.30 | 1 | | 04/05/24 17:43 | 75-01-4 | |
| Xylene (Total) | **<0.61** | mg/kg | 0.61 | 1 | | 04/05/24 17:43 | 1330-20-7 | |
| ***Surrogates*** | | | | | | | | |
| Dibromofluoromethane (S) | 97 | %. | 75-135 | 1 | | 04/05/24 17:43 | 1868-53-7 | |
| Toluene-d8 (S) | 100 | %. | 65-148 | 1 | | 04/05/24 17:43 | 2037-26-5 | |
| 4-Bromofluorobenzene (S) | 90 | %. | 63-132 | 1 | | 04/05/24 17:43 | 460-00-4 | |
| **8260 MSV TCLP** | Analytical Method: EPA 5030/8260 Leachate Method/Date: EPA 1311; 04/07/24 22:45 | | | | | | | |
| | Pace Analytical Services - Indianapolis | | | | | | | |
| Benzene | **<0.050** | mg/L | 0.050 | 1 | | 04/09/24 15:15 | 71-43-2 | |
| 2-Butanone (MEK) | **<1.0** | mg/L | 1.0 | 1 | | 04/09/24 15:15 | 78-93-3 | |
| Carbon tetrachloride | **<0.050** | mg/L | 0.050 | 1 | | 04/09/24 15:15 | 56-23-5 | |
| Chlorobenzene | **<0.050** | mg/L | 0.050 | 1 | | 04/09/24 15:15 | 108-90-7 | |
| Chloroform | **<0.050** | mg/L | 0.050 | 1 | | 04/09/24 15:15 | 67-66-3 | |
| 1,2-Dichloroethane | **<0.050** | mg/L | 0.050 | 1 | | 04/09/24 15:15 | 107-06-2 | |
| 1,1-Dichloroethene | **<0.050** | mg/L | 0.050 | 1 | | 04/09/24 15:15 | 75-35-4 | |
| Tetrachloroethene | **<0.050** | mg/L | 0.050 | 1 | | 04/09/24 15:15 | 127-18-4 | |
| Trichloroethene | **<0.050** | mg/L | 0.050 | 1 | | 04/09/24 15:15 | 79-01-6 | |
| Vinyl chloride | **<0.020** | mg/L | 0.020 | 1 | | 04/09/24 15:15 | 75-01-4 | |
| ***Surrogates*** | | | | | | | | |
| 4-Bromofluorobenzene (S) | 98 | %. | 79-124 | 1 | | 04/09/24 15:15 | 460-00-4 | |
| Dibromofluoromethane (S) | 107 | %. | 82-128 | 1 | | 04/09/24 15:15 | 1868-53-7 | |
| Toluene-d8 (S) | 98 | %. | 73-122 | 1 | | 04/09/24 15:15 | 2037-26-5 | |

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# ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50369587

**Sample: DIGESTOR #4** **Lab ID: 50369587001** Collected: 04/01/24 14:15 Received: 04/02/24 10:04 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **Percent Moisture** | Analytical Method: SM 2540G<br>Pace Analytical Services - Indianapolis | | | | | | | |
| Percent Moisture | **98.4** | % | 0.10 | 1 | | 04/04/24 16:37 | | N2 |
| **2540G Total Percent Solids** | Analytical Method: SM 2540G<br>Pace Analytical Services - Indianapolis | | | | | | | |
| Total Solids | **1.6** | % | 0.10 | 1 | | 04/04/24 16:38 | | N2 |
| **2540G Total Volatile Solids** | Analytical Method: SM 2540G<br>Pace Analytical Services - Indianapolis | | | | | | | |
| Total Volatile Solids | **58.8** | % | 0.10 | 1 | | 04/04/24 16:32 | | N2 |
| **9038 Sulfate Soil** | Analytical Method: EPA 9038 Preparation Method: EPA 9038<br>Pace Analytical Services - Indianapolis | | | | | | | |
| Sulfate | **48400** | mg/kg | 30500 | 5 | 04/08/24 11:08 | 04/08/24 13:45 | 14808-79-8 | N2 |
| **9045 pH Soil** | Analytical Method: EPA 9045<br>Pace Analytical Services - Indianapolis | | | | | | | |
| pH at 25 Degrees C | **8.4** | Std. Units | 0.10 | 1 | | 04/10/24 12:52 | | H3 |
| **351.2 Total Kjeldahl Nitrogen** | Analytical Method: EPA 351.2 Preparation Method: EPA 351.2<br>Pace Analytical Services - Indianapolis | | | | | | | |
| Nitrogen, Kjeldahl, Total | **115000** | mg/kg | 5750 | 2 | 04/05/24 11:51 | 04/06/24 14:07 | 7727-37-9 | N2 |
| **353.2 Nitrogen, NO2/NO3** | Analytical Method: EPA 353.2 Preparation Method: EPA 353.2<br>Pace Analytical Services - Indianapolis | | | | | | | |
| Nitrogen, NO2 plus NO3 | **592** | mg/kg | 302 | 1 | 04/10/24 02:56 | 04/11/24 20:33 | | N2 |
| Nitrogen, Nitrate | **<302** | mg/kg | 302 | 1 | 04/10/24 02:56 | 04/11/24 20:33 | 14797-55-8 | N2 |
| **4500 Ammonia Soil, Distilled** | Analytical Method: SM 4500-NH3 G Preparation Method: SM 4500-NH3 B<br>Pace Analytical Services - Indianapolis | | | | | | | |
| Nitrogen, Ammonia | **46500** | mg/kg | 2830 | 1 | 04/16/24 11:16 | 04/16/24 15:02 | 7664-41-7 | N2 |
| **4500 Chloride in Soil** | Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E<br>Pace Analytical Services - Indianapolis | | | | | | | |
| Chloride | **10300** | mg/kg | 6030 | 1 | 04/10/24 02:56 | 04/15/24 14:29 | 16887-00-6 | N2 |
| **4500CNE Cyanide, Soil** | Analytical Method: SM 4500-CN-E Preparation Method: SM 4500-CN-C<br>Pace Analytical Services - Indianapolis | | | | | | | |
| Cyanide | **<29.6** | mg/kg | 29.6 | 1 | 04/15/24 10:31 | 04/15/24 13:55 | 57-12-5 | N2 |

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 784349
QC Batch Method: EPA 7470
Analysis Method: EPA 7470
Analysis Description: 7470 Mercury TCLP
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

METHOD BLANK: 3588256 Matrix: Water
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Mercury | mg/L | <0.00067 | 0.00067 | 04/11/24 08:57 | |

LABORATORY CONTROL SAMPLE: 3588257

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Mercury | mg/L | 0.005 | 0.0048 | 96 | 80-120 | |

MATRIX SPIKE SAMPLE: 3588258

| Parameter | Units | 50369645001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Mercury | mg/L | ND | 0.015 | 0.014 | 91 | 75-125 | |

MATRIX SPIKE SAMPLE: 3588259

| Parameter | Units | 50369646001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Mercury | mg/L | ND | 0.015 | <0.020 | 85 | 75-125 | |

MATRIX SPIKE SAMPLE: 3588260

| Parameter | Units | 50369647001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Mercury | mg/L | ND | 0.015 | 0.014 | 94 | 75-125 | |

MATRIX SPIKE SAMPLE: 3588261

| Parameter | Units | 50369649001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Mercury | mg/L | ND | 0.015 | 0.014 | 93 | 75-125 | |

MATRIX SPIKE SAMPLE: 3588262

| Parameter | Units | 50369650001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Mercury | mg/L | ND | 0.015 | 0.014 | 92 | 75-125 | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

MATRIX SPIKE SAMPLE: 3588263

| Parameter | Units | 50369657001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Mercury | mg/L | ND | 0.15 | <0.020 | 9 | 75-125 | M0 |

MATRIX SPIKE SAMPLE: 3588264

| Parameter | Units | 50369577001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Mercury | mg/L | ND | 0.015 | 0.014 | 95 | 75-125 | |

MATRIX SPIKE SAMPLE: 3588265

| Parameter | Units | 50369581001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Mercury | mg/L | ND | 0.015 | 0.014 | 91 | 75-125 | |

MATRIX SPIKE SAMPLE: 3588266

| Parameter | Units | 50369582001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Mercury | mg/L | ND | 0.015 | 0.014 | 94 | 75-125 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3588267 3588268

| Parameter | Units | 50369587001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mercury | mg/L | <0.0020 | 0.015 | 0.015 | 0.014 | 0.015 | 95 | 97 | 75-125 | 2 | 20 | |

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**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 784384
QC Batch Method: EPA 7471
Analysis Method: EPA 7471
Analysis Description: 7471 Mercury
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

METHOD BLANK: 3588424 Matrix: Solid
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Mercury | mg/kg | <0.20 | 0.20 | 04/11/24 18:33 | |

LABORATORY CONTROL SAMPLE: 3588425

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Mercury | mg/kg | 0.5 | 0.52 | 104 | 80-120 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3588426 3588427

| Parameter | Units | 50369758002 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mercury | mg/kg | ND | 0.62 | 0.63 | 0.67 | 0.70 | 104 | 107 | 75-125 | 4 | 20 | |

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## REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 783830
QC Batch Method: EPA 3050
Analysis Method: EPA 6010
Analysis Description: 6010 MET
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

METHOD BLANK: 3586533 Matrix: Solid
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Aluminum | mg/kg | <50.0 | 50.0 | 04/10/24 11:56 | |
| Calcium | mg/kg | <50.0 | 50.0 | 04/10/24 11:56 | |
| Magnesium | mg/kg | <50.0 | 50.0 | 04/10/24 11:56 | |
| Phosphorus | mg/kg | ND | 50.0 | 04/10/24 11:56 | N2 |
| Potassium | mg/kg | <50.0 | 50.0 | 04/10/24 11:56 | |
| Sodium | mg/kg | <50.0 | 50.0 | 04/10/24 11:56 | |

LABORATORY CONTROL SAMPLE: 3586534

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Aluminum | mg/kg | 500 | 488 | 98 | 80-120 | |
| Calcium | mg/kg | 500 | 495 | 99 | 80-120 | |
| Magnesium | mg/kg | 500 | 470 | 94 | 80-120 | |
| Phosphorus | mg/kg | 50 | <50.0 | 98 | 80-120 | N2 |
| Potassium | mg/kg | 500 | 474 | 95 | 80-120 | |
| Sodium | mg/kg | 500 | 470 | 94 | 80-120 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3586535 3586536

| Parameter | Units | 50369511008 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Aluminum | mg/kg | 9630 | 532 | 504 | 12300 | 11900 | 506 | 458 | 75-125 | 3 | 20 | P6 |
| Calcium | mg/kg | 28400 | 532 | 504 | 33500 | 34400 | 958 | 1180 | 75-125 | 2 | 20 | E,P6 |
| Magnesium | mg/kg | 6490 | 532 | 504 | 7630 | 7540 | 214 | 209 | 75-125 | 1 | 20 | P6 |
| Phosphorus | mg/kg | 353 | 53.2 | 50.4 | 426 | 411 | 138 | 117 | 75-125 | 4 | 20 | M0,N2 |
| Potassium | mg/kg | 1660 | 532 | 504 | 2860 | 2760 | 227 | 219 | 75-125 | 4 | 20 | M3 |
| Sodium | mg/kg | 90.1 | 532 | 504 | 575 | 538 | 91 | 89 | 75-125 | 6 | 20 | |

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 784153
QC Batch Method: EPA 3010
Analysis Method: EPA 6010
Analysis Description: 6010 MET TCLP
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

METHOD BLANK: 3587555 Matrix: Water
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Arsenic | mg/L | <0.010 | 0.010 | 04/10/24 09:30 | |
| Barium | mg/L | <0.50 | 0.50 | 04/10/24 09:30 | |
| Cadmium | mg/L | <0.0050 | 0.0050 | 04/10/24 09:30 | |
| Chromium | mg/L | <0.010 | 0.010 | 04/10/24 09:30 | |
| Lead | mg/L | <0.010 | 0.010 | 04/10/24 09:30 | |
| Selenium | mg/L | <0.010 | 0.010 | 04/10/24 09:30 | |
| Silver | mg/L | <0.010 | 0.010 | 04/10/24 09:30 | |
| Zinc | mg/L | <0.020 | 0.020 | 04/10/24 09:30 | |

LABORATORY CONTROL SAMPLE: 3587556

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Arsenic | mg/L | 1 | 0.97 | 97 | 80-120 | |
| Barium | mg/L | 1 | 0.96 | 96 | 80-120 | |
| Cadmium | mg/L | 1 | 0.95 | 95 | 80-120 | |
| Chromium | mg/L | 1 | 0.99 | 99 | 80-120 | |
| Lead | mg/L | 1 | 0.93 | 93 | 80-120 | |
| Selenium | mg/L | 1 | 0.96 | 96 | 80-120 | F5 |
| Silver | mg/L | 0.5 | 0.46 | 92 | 80-120 | |
| Zinc | mg/L | 1 | 0.98 | 98 | 80-120 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3587557 3587558

| Parameter | Units | 50367901004 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Arsenic | mg/L | <0.050 | 10 | 10 | 9.7 | 9.4 | 97 | 94 | 50-150 | 3 | 20 | |
| Barium | mg/L | 1.1J | 10 | 10 | 10.6 | 10.3 | 95 | 91 | 50-150 | 4 | 20 | |
| Cadmium | mg/L | <0.025 | 10 | 10 | 9.5 | 9.2 | 95 | 92 | 50-150 | 3 | 20 | |
| Chromium | mg/L | <0.052 | 10 | 10 | 9.7 | 9.4 | 97 | 94 | 50-150 | 3 | 20 | |
| Lead | mg/L | <0.050 | 10 | 10 | 8.9 | 8.7 | 89 | 87 | 50-150 | 3 | 20 | |
| Selenium | mg/L | <0.050 | 10 | 10 | 9.6 | 9.4 | 96 | 94 | 50-150 | 3 | 20 | F5 |
| Silver | mg/L | <0.050 | 5 | 5 | 4.6 | 4.4 | 91 | 88 | 50-150 | 3 | 20 | |
| Zinc | mg/L | 0.86 | 10 | 10 | 10.5 | 10.2 | 97 | 94 | 50-150 | 3 | 20 | |

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

MATRIX SPIKE SAMPLE: 3587559

| Parameter | Units | 50368965005 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Arsenic | mg/L | ND | 10 | 9.5 | 95 | 50-150 | |
| Barium | mg/L | ND | 10 | 10.6 | 93 | 50-150 | |
| Cadmium | mg/L | ND | 10 | 9.4 | 94 | 50-150 | |
| Chromium | mg/L | ND | 10 | 9.6 | 96 | 50-150 | |
| Lead | mg/L | ND | 10 | 8.9 | 89 | 50-150 | |
| Selenium | mg/L | ND | 10 | 9.4 | 94 | 50-150 | F5 |
| Silver | mg/L | ND | 5 | 4.5 | 90 | 50-150 | |
| Zinc | mg/L | 387 ug/L | 10 | 10 | 96 | 50-150 | |

MATRIX SPIKE SAMPLE: 3587560

| Parameter | Units | 50369657001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Arsenic | mg/L | ND | 50 | 49.4 | 99 | 50-150 | |
| Barium | mg/L | ND | 50 | 50.5 | 100 | 50-150 | |
| Cadmium | mg/L | ND | 50 | 49.1 | 98 | 50-150 | |
| Chromium | mg/L | ND | 50 | 51.1 | 102 | 50-150 | |
| Lead | mg/L | ND | 50 | 47.5 | 95 | 50-150 | |
| Selenium | mg/L | ND | 50 | 49.1 | 98 | 50-150 | F5 |
| Silver | mg/L | ND | 25 | 23.5 | 94 | 50-150 | |
| Zinc | mg/L | ND | 50 | 52.3 | 101 | 50-150 | |

MATRIX SPIKE SAMPLE: 3587561

| Parameter | Units | 50369577001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Arsenic | mg/L | ND | 10 | 9.7 | 97 | 50-150 | |
| Barium | mg/L | ND | 10 | 9.8 | 94 | 50-150 | |
| Cadmium | mg/L | ND | 10 | 9.4 | 94 | 50-150 | |
| Chromium | mg/L | ND | 10 | 9.7 | 97 | 50-150 | |
| Lead | mg/L | ND | 10 | 8.9 | 89 | 50-150 | |
| Selenium | mg/L | ND | 10 | 9.6 | 96 | 50-150 | F5 |
| Silver | mg/L | ND | 5 | 4.6 | 91 | 50-150 | |
| Zinc | mg/L | ND | 10 | 9.7 | 96 | 50-150 | |

MATRIX SPIKE SAMPLE: 3587562

| Parameter | Units | 50369581001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Arsenic | mg/L | ND | 10 | 9.5 | 95 | 50-150 | |
| Barium | mg/L | ND | 10 | 10.1 | 94 | 50-150 | |
| Cadmium | mg/L | ND | 10 | 9.3 | 93 | 50-150 | |
| Chromium | mg/L | ND | 10 | 9.5 | 95 | 50-150 | |
| Lead | mg/L | ND | 10 | 8.8 | 88 | 50-150 | |
| Selenium | mg/L | ND | 10 | 9.5 | 94 | 50-150 | F5 |

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## QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

MATRIX SPIKE SAMPLE: 3587562

| Parameter | Units | 50369581001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Silver | mg/L | ND | 5 | 4.5 | 90 | 50-150 | |
| Zinc | mg/L | ND | 10 | 9.6 | 95 | 50-150 | |

MATRIX SPIKE SAMPLE: 3587563

| Parameter | Units | 50369582001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Arsenic | mg/L | ND | 10 | 9.4 | 94 | 50-150 | |
| Barium | mg/L | ND | 10 | 9.7 | 91 | 50-150 | |
| Cadmium | mg/L | ND | 10 | 9.2 | 92 | 50-150 | |
| Chromium | mg/L | ND | 10 | 9.4 | 94 | 50-150 | |
| Lead | mg/L | ND | 10 | 8.6 | 86 | 50-150 | |
| Selenium | mg/L | ND | 10 | 9.3 | 92 | 50-150 | F5 |
| Silver | mg/L | ND | 5 | 4.4 | 88 | 50-150 | |
| Zinc | mg/L | ND | 10 | 9.4 | 94 | 50-150 | |

MATRIX SPIKE SAMPLE: 3587564

| Parameter | Units | 50369587001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Arsenic | mg/L | <0.10 | 10 | 10 | 100 | 50-150 | |
| Barium | mg/L | <5.0 | 10 | 9.8 | 97 | 50-150 | |
| Cadmium | mg/L | <0.050 | 10 | 9.8 | 97 | 50-150 | |
| Chromium | mg/L | <0.10 | 10 | 10.0 | 100 | 50-150 | |
| Lead | mg/L | <0.10 | 10 | 9.3 | 93 | 50-150 | |
| Selenium | mg/L | <0.10 | 10 | 9.9 | 98 | 50-150 | F5 |
| Silver | mg/L | <0.10 | 5 | 4.7 | 94 | 50-150 | |
| Zinc | mg/L | 3.9 | 10 | 14.2 | 102 | 50-150 | |

MATRIX SPIKE SAMPLE: 3587565

| Parameter | Units | 50370084002 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Arsenic | mg/L | ND | 10 | 9.7 | 97 | 50-150 | |
| Barium | mg/L | ND | 10 | 9.8 | 95 | 50-150 | |
| Cadmium | mg/L | ND | 10 | 9.4 | 94 | 50-150 | |
| Chromium | mg/L | ND | 10 | 9.7 | 97 | 50-150 | |
| Lead | mg/L | ND | 10 | 9.0 | 90 | 50-150 | |
| Selenium | mg/L | ND | 10 | 9.6 | 96 | 50-150 | F5 |
| Silver | mg/L | ND | 5 | 4.5 | 90 | 50-150 | |
| Zinc | mg/L | 326 ug/L | 10 | 10.1 | 97 | 50-150 | |

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**Pace Analytical Services, LLC**
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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 784173
QC Batch Method: EPA 3050B
Analysis Method: EPA 6020
Analysis Description: 6020 MET
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

METHOD BLANK: 3587647 Matrix: Solid

Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Arsenic | mg/kg | <0.10 | 0.10 | 04/12/24 03:15 | |
| Cadmium | mg/kg | <0.050 | 0.050 | 04/12/24 03:15 | |
| Chromium | mg/kg | <0.20 | 0.20 | 04/12/24 03:15 | |
| Copper | mg/kg | <0.10 | 0.10 | 04/12/24 03:15 | |
| Lead | mg/kg | <0.10 | 0.10 | 04/12/24 03:15 | |
| Molybdenum | mg/kg | <0.10 | 0.10 | 04/12/24 03:15 | N2 |
| Nickel | mg/kg | <0.10 | 0.10 | 04/12/24 03:15 | |
| Selenium | mg/kg | <0.10 | 0.10 | 04/12/24 03:15 | |
| Silver | mg/kg | <0.050 | 0.050 | 04/12/24 03:15 | |
| Zinc | mg/kg | <0.50 | 0.50 | 04/12/24 03:15 | |

LABORATORY CONTROL SAMPLE: 3587648

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Arsenic | mg/kg | 4 | 3.9 | 97 | 80-120 | |
| Cadmium | mg/kg | 4 | 4.1 | 103 | 80-120 | |
| Chromium | mg/kg | 4 | 4.2 | 104 | 80-120 | |
| Copper | mg/kg | 4 | 4.1 | 102 | 80-120 | |
| Lead | mg/kg | 4 | 4.1 | 102 | 80-120 | |
| Molybdenum | mg/kg | 4 | 4.0 | 101 | 80-120 | N2 |
| Nickel | mg/kg | 4 | 4.1 | 102 | 80-120 | |
| Selenium | mg/kg | 4 | 4.0 | 100 | 80-120 | |
| Silver | mg/kg | 4 | 4.1 | 101 | 80-120 | |
| Zinc | mg/kg | 4 | 4.1 | 103 | 80-120 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3587649 3587650

| Parameter | Units | 50369587001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Arsenic | mg/kg | 18.2 | 244 | 244 | 245 | 245 | 94 | 94 | 75-125 | 0 | 20 | |
| Cadmium | mg/kg | 7.3 | 244 | 244 | 255 | 252 | 103 | 101 | 75-125 | 1 | 20 | |
| Chromium | mg/kg | 123 | 244 | 244 | 367 | 374 | 101 | 104 | 75-125 | 2 | 20 | |
| Copper | mg/kg | 818 | 244 | 244 | 1060 | 1070 | 101 | 103 | 75-125 | 1 | 20 | |
| Lead | mg/kg | 16.6 | 244 | 244 | 231 | 230 | 89 | 88 | 75-125 | 0 | 20 | |
| Molybdenum | mg/kg | 21.0 | 244 | 244 | 262 | 265 | 100 | 101 | 75-125 | 1 | 20 | N2 |
| Nickel | mg/kg | 26.5 | 244 | 244 | 265 | 266 | 99 | 99 | 75-125 | 0 | 20 | |
| Selenium | mg/kg | <6.0 | 244 | 244 | 240 | 240 | 97 | 97 | 75-125 | 0 | 20 | |

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3587649 3587650

| Parameter | Units | 50369587001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Silver | mg/kg | 4.7 | 244 | 244 | 242 | 243 | 98 | 99 | 75-125 | 1 | 20 | |
| Zinc | mg/kg | 2420 | 244 | 244 | 2580 | 2610 | 64 | 76 | 75-125 | 1 | 20 | P6 |

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 783728
QC Batch Method: EPA 8260
Analysis Method: EPA 8260
Analysis Description: 8260 MSV 5030 Low
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50369587001

METHOD BLANK: 3585889 Matrix: Solid
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| 1,1,1,2-Tetrachloroethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 1,1,1-Trichloroethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 1,1,2,2-Tetrachloroethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 1,1,2-Trichloroethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 1,1-Dichloroethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 1,1-Dichloroethene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 1,2,3-Trichloropropane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 1,2-Dibromo-3-chloropropane | mg/kg | <0.010 | 0.010 | 04/05/24 12:40 | |
| 1,2-Dibromoethane (EDB) | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 1,2-Dichlorobenzene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 1,2-Dichloroethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 1,2-Dichloropropane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 1,3-Dichlorobenzene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 1,4-Dichlorobenzene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| 2-Butanone (MEK) | mg/kg | <0.025 | 0.025 | 04/05/24 12:40 | |
| 2-Hexanone | mg/kg | <0.10 | 0.10 | 04/05/24 12:40 | |
| 4-Methyl-2-pentanone (MIBK) | mg/kg | <0.025 | 0.025 | 04/05/24 12:40 | |
| Acetone | mg/kg | <0.10 | 0.10 | 04/05/24 12:40 | |
| Acrolein | mg/kg | <0.10 | 0.10 | 04/05/24 12:40 | |
| Acrylonitrile | mg/kg | <0.10 | 0.10 | 04/05/24 12:40 | |
| Benzene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Bromodichloromethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Bromoform | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Bromomethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Carbon disulfide | mg/kg | <0.010 | 0.010 | 04/05/24 12:40 | |
| Carbon tetrachloride | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Chlorobenzene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Chloroethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Chloroform | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Chloromethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| cis-1,2-Dichloroethene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| cis-1,3-Dichloropropene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Dibromochloromethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Dibromomethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Dichlorodifluoromethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Ethyl methacrylate | mg/kg | <0.10 | 0.10 | 04/05/24 12:40 | |
| Ethylbenzene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Iodomethane | mg/kg | <0.10 | 0.10 | 04/05/24 12:40 | |
| Methyl methacrylate | mg/kg | <0.10 | 0.10 | 04/05/24 12:40 | |
| Methyl-tert-butyl ether | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |

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## REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

METHOD BLANK: 3585889 Matrix: Solid
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Methylene Chloride | mg/kg | <0.020 | 0.020 | 04/05/24 12:40 | |
| Styrene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Tetrachloroethene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Toluene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| trans-1,2-Dichloroethene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| trans-1,3-Dichloropropene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| trans-1,4-Dichloro-2-butene | mg/kg | <0.10 | 0.10 | 04/05/24 12:40 | |
| Trichloroethene | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Trichlorofluoromethane | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Vinyl acetate | mg/kg | <0.10 | 0.10 | 04/05/24 12:40 | |
| Vinyl chloride | mg/kg | <0.0050 | 0.0050 | 04/05/24 12:40 | |
| Xylene (Total) | mg/kg | <0.010 | 0.010 | 04/05/24 12:40 | |
| 4-Bromofluorobenzene (S) | %. | 97 | 63-132 | 04/05/24 12:40 | |
| Dibromofluoromethane (S) | %. | 99 | 75-135 | 04/05/24 12:40 | |
| Toluene-d8 (S) | %. | 95 | 65-148 | 04/05/24 12:40 | |

LABORATORY CONTROL SAMPLE: 3585890

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| 1,1,1,2-Tetrachloroethane | mg/kg | 0.05 | 0.052 | 103 | 70-129 | |
| 1,1,1-Trichloroethane | mg/kg | 0.05 | 0.051 | 102 | 67-134 | |
| 1,1,2,2-Tetrachloroethane | mg/kg | 0.05 | 0.044 | 88 | 67-122 | |
| 1,1,2-Trichloroethane | mg/kg | 0.05 | 0.045 | 91 | 72-127 | |
| 1,1-Dichloroethane | mg/kg | 0.05 | 0.045 | 90 | 72-121 | |
| 1,1-Dichloroethene | mg/kg | 0.05 | 0.057 | 114 | 57-140 | |
| 1,2,3-Trichloropropane | mg/kg | 0.05 | 0.048 | 96 | 70-124 | |
| 1,2-Dibromo-3-chloropropane | mg/kg | 0.05 | 0.051 | 102 | 56-127 | |
| 1,2-Dibromoethane (EDB) | mg/kg | 0.05 | 0.049 | 99 | 71-126 | |
| 1,2-Dichlorobenzene | mg/kg | 0.05 | 0.049 | 99 | 68-120 | |
| 1,2-Dichloroethane | mg/kg | 0.05 | 0.046 | 93 | 67-129 | |
| 1,2-Dichloropropane | mg/kg | 0.05 | 0.044 | 88 | 71-123 | |
| 1,3-Dichlorobenzene | mg/kg | 0.05 | 0.050 | 99 | 65-121 | |
| 1,4-Dichlorobenzene | mg/kg | 0.05 | 0.050 | 100 | 66-122 | |
| 2-Butanone (MEK) | mg/kg | 0.25 | 0.21 | 83 | 59-136 | |
| 2-Hexanone | mg/kg | 0.25 | 0.21 | 82 | 62-127 | |
| 4-Methyl-2-pentanone (MIBK) | mg/kg | 0.25 | 0.20 | 82 | 67-131 | |
| Acetone | mg/kg | 0.25 | 0.26 | 102 | 45-127 | |
| Acrolein | mg/kg | 1 | 1.1 | 113 | 42-158 | |
| Acrylonitrile | mg/kg | 0.25 | 0.21 | 86 | 69-127 | |
| Benzene | mg/kg | 0.05 | 0.047 | 94 | 69-125 | |
| Bromodichloromethane | mg/kg | 0.05 | 0.050 | 101 | 77-130 | |
| Bromoform | mg/kg | 0.05 | 0.051 | 101 | 67-128 | |
| Bromomethane | mg/kg | 0.05 | 0.080 | 161 | 60-156 | L1 |
| Carbon disulfide | mg/kg | 0.05 | 0.053 | 106 | 47-137 | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

## REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

LABORATORY CONTROL SAMPLE: 3585890

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Carbon tetrachloride | mg/kg | 0.05 | 0.052 | 105 | 68-132 | |
| Chlorobenzene | mg/kg | 0.05 | 0.048 | 96 | 68-122 | |
| Chloroethane | mg/kg | 0.05 | 0.060 | 121 | 61-137 | |
| Chloroform | mg/kg | 0.05 | 0.049 | 99 | 71-124 | |
| Chloromethane | mg/kg | 0.05 | 0.042 | 84 | 56-131 | |
| cis-1,2-Dichloroethene | mg/kg | 0.05 | 0.049 | 98 | 70-123 | |
| cis-1,3-Dichloropropene | mg/kg | 0.05 | 0.048 | 97 | 72-136 | |
| Dibromochloromethane | mg/kg | 0.05 | 0.051 | 102 | 73-130 | |
| Dibromomethane | mg/kg | 0.05 | 0.049 | 98 | 74-123 | |
| Dichlorodifluoromethane | mg/kg | 0.05 | 0.020 | 40 | 23-127 | |
| Ethyl methacrylate | mg/kg | 0.05 | <0.10 | 99 | 70-131 | |
| Ethylbenzene | mg/kg | 0.05 | 0.049 | 99 | 65-124 | |
| Iodomethane | mg/kg | 0.05 | <0.10 | 105 | 50-137 | |
| Methyl methacrylate | mg/kg | 0.05 | <0.10 | 91 | 63-137 | |
| Methyl-tert-butyl ether | mg/kg | 0.05 | 0.048 | 95 | 69-128 | |
| Methylene Chloride | mg/kg | 0.05 | 0.050 | 100 | 61-128 | |
| Styrene | mg/kg | 0.05 | 0.050 | 99 | 68-124 | |
| Tetrachloroethene | mg/kg | 0.05 | 0.051 | 102 | 62-128 | |
| Toluene | mg/kg | 0.05 | 0.048 | 95 | 60-122 | |
| trans-1,2-Dichloroethene | mg/kg | 0.05 | 0.048 | 97 | 67-124 | |
| trans-1,3-Dichloropropene | mg/kg | 0.05 | 0.050 | 99 | 68-136 | |
| trans-1,4-Dichloro-2-butene | mg/kg | 0.05 | <0.10 | 89 | 64-134 | |
| Trichloroethene | mg/kg | 0.05 | 0.049 | 97 | 68-128 | |
| Trichlorofluoromethane | mg/kg | 0.05 | 0.054 | 108 | 57-146 | |
| Vinyl acetate | mg/kg | 0.2 | 0.20 | 101 | 56-181 | |
| Vinyl chloride | mg/kg | 0.05 | 0.048 | 96 | 52-142 | |
| Xylene (Total) | mg/kg | 0.15 | 0.14 | 95 | 62-122 | |
| 4-Bromofluorobenzene (S) | %. | | | 96 | 63-132 | |
| Dibromofluoromethane (S) | %. | | | 99 | 75-135 | |
| Toluene-d8 (S) | %. | | | 96 | 65-148 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3585891 3585892

| Parameter | Units | 50369747001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,1,1,2-Tetrachloroethane | mg/kg | ND | 0.046 | 0.047 | 0.039 | 0.045 | 85 | 95 | 22-160 | 12 | 20 | |
| 1,1,1-Trichloroethane | mg/kg | ND | 0.046 | 0.047 | 0.041 | 0.047 | 89 | 100 | 52-148 | 13 | 20 | |
| 1,1,2,2-Tetrachloroethane | mg/kg | ND | 0.046 | 0.047 | 0.037 | 0.041 | 80 | 88 | 24-166 | 12 | 20 | |
| 1,1,2-Trichloroethane | mg/kg | ND | 0.046 | 0.047 | 0.036 | 0.040 | 78 | 86 | 30-162 | 12 | 20 | |
| 1,1-Dichloroethane | mg/kg | ND | 0.046 | 0.047 | 0.037 | 0.042 | 79 | 89 | 49-138 | 14 | 20 | |
| 1,1-Dichloroethene | mg/kg | ND | 0.046 | 0.047 | 0.061 | 0.070 | 131 | 148 | 39-162 | 14 | 20 | |
| 1,2,3-Trichloropropane | mg/kg | ND | 0.046 | 0.047 | 0.040 | 0.045 | 87 | 97 | 17-177 | 12 | 20 | |
| 1,2-Dibromo-3-chloropropane | mg/kg | ND | 0.046 | 0.047 | 0.035 | 0.040 | 76 | 86 | 10-148 | 14 | 20 | |
| 1,2-Dibromoethane (EDB) | mg/kg | ND | 0.046 | 0.047 | 0.036 | 0.040 | 77 | 85 | 36-141 | 12 | 20 | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

## REPORT OF LABORATORY ANALYSIS

**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

## QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3585891 3585892

| Parameter | Units | 50369747001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,2-Dichlorobenzene | mg/kg | ND | 0.046 | 0.047 | 0.035 | 0.037 | 75 | 79 | 10-136 | 7 | 20 | |
| 1,2-Dichloroethane | mg/kg | ND | 0.046 | 0.047 | 0.033 | 0.038 | 72 | 82 | 48-138 | 14 | 20 | |
| 1,2-Dichloropropane | mg/kg | ND | 0.046 | 0.047 | 0.035 | 0.040 | 76 | 84 | 45-140 | 12 | 20 | |
| 1,3-Dichlorobenzene | mg/kg | ND | 0.046 | 0.047 | 0.036 | 0.039 | 78 | 84 | 10-135 | 9 | 20 | |
| 1,4-Dichlorobenzene | mg/kg | ND | 0.046 | 0.047 | 0.037 | 0.041 | 80 | 86 | 10-136 | 9 | 20 | |
| 2-Butanone (MEK) | mg/kg | ND | 0.23 | 0.24 | 0.12 | 0.13 | 52 | 55 | 33-160 | 7 | 20 | |
| 2-Hexanone | mg/kg | ND | 0.23 | 0.24 | 0.10 | 0.10 | 44 | 43 | 18-155 | 1 | 20 | |
| 4-Methyl-2-pentanone (MIBK) | mg/kg | ND | 0.23 | 0.24 | 0.13 | 0.14 | 55 | 59 | 27-175 | 9 | 20 | |
| Acetone | mg/kg | ND | 0.23 | 0.24 | 0.48 | 0.48 | 169 | 163 | 18-159 | 2 | 20 | M1 |
| Acrolein | mg/kg | ND | 0.92 | 0.94 | <0.092 | <0.094 | 3 | 2 | 10-155 | | 20 | M1 |
| Acrylonitrile | mg/kg | ND | 0.23 | 0.24 | <0.092 | <0.094 | 22 | 15 | 24-157 | | 20 | M1 |
| Benzene | mg/kg | ND | 0.046 | 0.047 | 0.038 | 0.044 | 82 | 93 | 48-137 | 14 | 20 | |
| Bromodichloromethane | mg/kg | ND | 0.046 | 0.047 | 0.034 | 0.039 | 74 | 83 | 32-152 | 13 | 20 | |
| Bromoform | mg/kg | ND | 0.046 | 0.047 | 0.035 | 0.039 | 76 | 82 | 10-178 | 9 | 20 | |
| Bromomethane | mg/kg | ND | 0.046 | 0.047 | 0.060 | 0.064 | 129 | 136 | 31-164 | 7 | 20 | |
| Carbon disulfide | mg/kg | ND | 0.046 | 0.047 | 0.048 | 0.058 | 101 | 122 | 23-145 | 20 | 20 | |
| Carbon tetrachloride | mg/kg | ND | 0.046 | 0.047 | 0.033 | 0.036 | 71 | 77 | 43-148 | 10 | 20 | |
| Chlorobenzene | mg/kg | ND | 0.046 | 0.047 | 0.039 | 0.044 | 85 | 93 | 28-136 | 11 | 20 | |
| Chloroethane | mg/kg | ND | 0.046 | 0.047 | 0.058 | 0.075 | 126 | 160 | 34-160 | 25 | 20 | R1 |
| Chloroform | mg/kg | ND | 0.046 | 0.047 | 0.041 | 0.046 | 88 | 98 | 54-136 | 13 | 20 | |
| Chloromethane | mg/kg | ND | 0.046 | 0.047 | 0.038 | 0.042 | 82 | 90 | 36-145 | 12 | 20 | |
| cis-1,2-Dichloroethene | mg/kg | ND | 0.046 | 0.047 | 0.040 | 0.045 | 86 | 96 | 52-132 | 13 | 20 | |
| cis-1,3-Dichloropropene | mg/kg | ND | 0.046 | 0.047 | 0.028 | 0.029 | 61 | 61 | 22-163 | 2 | 20 | |
| Dibromochloromethane | mg/kg | ND | 0.046 | 0.047 | 0.033 | 0.037 | 72 | 79 | 18-161 | 11 | 20 | |
| Dibromomethane | mg/kg | ND | 0.046 | 0.047 | 0.036 | 0.041 | 79 | 88 | 32-147 | 13 | 20 | |
| Dichlorodifluoromethane | mg/kg | ND | 0.046 | 0.047 | 0.014 | 0.016 | 30 | 34 | 10-138 | 15 | 20 | |
| Ethyl methacrylate | mg/kg | ND | 0.046 | 0.047 | <0.092 | <0.094 | 3 | 1 | 10-167 | | 20 | M1 |
| Ethylbenzene | mg/kg | ND | 0.046 | 0.047 | 0.038 | 0.043 | 83 | 91 | 24-150 | 12 | 20 | |
| Iodomethane | mg/kg | ND | 0.046 | 0.047 | <0.092 | <0.094 | 82 | 86 | 23-142 | | 20 | |
| Methyl methacrylate | mg/kg | ND | 0.046 | 0.047 | <0.092 | <0.094 | 43 | 27 | 10-177 | | 20 | |
| Methyl-tert-butyl ether | mg/kg | ND | 0.046 | 0.047 | 0.034 | 0.039 | 71 | 82 | 57-141 | 15 | 20 | |
| Methylene Chloride | mg/kg | ND | 0.046 | 0.047 | 0.063 | 0.078 | 99 | 129 | 40-140 | 21 | 20 | R1 |
| Styrene | mg/kg | ND | 0.046 | 0.047 | 0.036 | 0.040 | 78 | 84 | 12-138 | 9 | 20 | |
| Tetrachloroethene | mg/kg | ND | 0.046 | 0.047 | 0.042 | 0.048 | 91 | 102 | 26-159 | 13 | 20 | |
| Toluene | mg/kg | ND | 0.046 | 0.047 | 0.042 | 0.047 | 90 | 97 | 28-150 | 10 | 20 | |
| trans-1,2-Dichloroethene | mg/kg | ND | 0.046 | 0.047 | 0.040 | 0.046 | 86 | 98 | 50-134 | 15 | 20 | |
| trans-1,3-Dichloropropene | mg/kg | ND | 0.046 | 0.047 | 0.029 | 0.031 | 64 | 67 | 17-153 | 7 | 20 | |
| trans-1,4-Dichloro-2-butene | mg/kg | ND | 0.046 | 0.047 | <0.092 | <0.094 | 33 | 24 | 10-146 | | 20 | |
| Trichloroethene | mg/kg | ND | 0.046 | 0.047 | 0.039 | 0.044 | 85 | 95 | 33-155 | 12 | 20 | |
| Trichlorofluoromethane | mg/kg | ND | 0.046 | 0.047 | 0.049 | 0.056 | 106 | 120 | 37-163 | 14 | 20 | |
| Vinyl acetate | mg/kg | ND | 0.18 | 0.19 | <0.092 | <0.094 | 15 | 17 | 10-183 | | 20 | |
| Vinyl chloride | mg/kg | ND | 0.046 | 0.047 | 0.045 | 0.050 | 98 | 106 | 37-161 | 10 | 20 | |
| Xylene (Total) | mg/kg | ND | 0.14 | 0.14 | 0.11 | 0.12 | 79 | 86 | 25-142 | 11 | 20 | |
| 4-Bromofluorobenzene (S) | %. | | | | | | 90 | 90 | 63-132 | | | |
| Dibromofluoromethane (S) | %. | | | | | | 96 | 96 | 75-135 | | | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3585891 3585892

| Parameter | Units | 50369747001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Toluene-d8 (S) | %. | | | | | | 100 | 100 | 65-148 | | | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3585893 3585894

| Parameter | Units | 50369748001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,1,1,2-Tetrachloroethane | mg/kg | ND | 0.047 | 0.049 | 0.036 | 0.036 | 77 | 74 | 22-160 | 2 | 20 | |
| 1,1,1-Trichloroethane | mg/kg | ND | 0.047 | 0.049 | 0.042 | 0.041 | 89 | 84 | 52-148 | 0 | 20 | |
| 1,1,2,2-Tetrachloroethane | mg/kg | ND | 0.047 | 0.049 | 0.031 | 0.032 | 67 | 64 | 24-166 | 2 | 20 | |
| 1,1,2-Trichloroethane | mg/kg | ND | 0.047 | 0.049 | 0.036 | 0.037 | 78 | 74 | 30-162 | 1 | 20 | |
| 1,1-Dichloroethane | mg/kg | ND | 0.047 | 0.049 | 0.039 | 0.039 | 84 | 79 | 49-138 | 0 | 20 | |
| 1,1-Dichloroethene | mg/kg | ND | 0.047 | 0.049 | 0.067 | 0.070 | 143 | 142 | 39-162 | 5 | 20 | |
| 1,2,3-Trichloropropane | mg/kg | ND | 0.047 | 0.049 | 0.035 | 0.035 | 75 | 72 | 17-177 | 2 | 20 | |
| 1,2-Dibromo-3-chloropropane | mg/kg | ND | 0.047 | 0.049 | 0.030 | 0.033 | 65 | 66 | 10-148 | 8 | 20 | |
| 1,2-Dibromoethane (EDB) | mg/kg | ND | 0.047 | 0.049 | 0.034 | 0.035 | 74 | 70 | 36-141 | 1 | 20 | |
| 1,2-Dichlorobenzene | mg/kg | ND | 0.047 | 0.049 | 0.026 | 0.026 | 56 | 53 | 10-136 | 0 | 20 | |
| 1,2-Dichloroethane | mg/kg | ND | 0.047 | 0.049 | 0.036 | 0.036 | 77 | 74 | 48-138 | 2 | 20 | |
| 1,2-Dichloropropane | mg/kg | ND | 0.047 | 0.049 | 0.037 | 0.037 | 79 | 76 | 45-140 | 2 | 20 | |
| 1,3-Dichlorobenzene | mg/kg | ND | 0.047 | 0.049 | 0.027 | 0.027 | 58 | 54 | 10-135 | 2 | 20 | |
| 1,4-Dichlorobenzene | mg/kg | ND | 0.047 | 0.049 | 0.028 | 0.027 | 60 | 55 | 10-136 | 3 | 20 | |
| 2-Butanone (MEK) | mg/kg | ND | 0.23 | 0.25 | 0.13 | 0.14 | 48 | 50 | 33-160 | 10 | 20 | |
| 2-Hexanone | mg/kg | ND | 0.23 | 0.25 | 0.095 | <0.099 | 41 | 33 | 18-155 | | 20 | |
| 4-Methyl-2-pentanone (MIBK) | mg/kg | ND | 0.23 | 0.25 | 0.13 | 0.12 | 57 | 48 | 27-175 | 9 | 20 | |
| Acetone | mg/kg | ND | 0.23 | 0.25 | 0.43 | 0.55 | 147 | 186 | 18-159 | 24 | 20 | M1,R1 |
| Acrolein | mg/kg | ND | 0.93 | 0.99 | <0.093 | <0.099 | 1 | 2 | 10-155 | | 20 | M1 |
| Acrylonitrile | mg/kg | ND | 0.23 | 0.25 | <0.093 | <0.099 | 9 | 12 | 24-157 | | 20 | M1 |
| Benzene | mg/kg | ND | 0.047 | 0.049 | 0.040 | 0.040 | 85 | 81 | 48-137 | 1 | 20 | |
| Bromodichloromethane | mg/kg | ND | 0.047 | 0.049 | 0.031 | 0.033 | 68 | 66 | 32-152 | 4 | 20 | |
| Bromoform | mg/kg | ND | 0.047 | 0.049 | 0.025 | 0.027 | 55 | 54 | 10-178 | 5 | 20 | |
| Bromomethane | mg/kg | ND | 0.047 | 0.049 | 0.060 | 0.063 | 130 | 128 | 31-164 | 5 | 20 | |
| Carbon disulfide | mg/kg | ND | 0.047 | 0.049 | 0.049 | 0.051 | 101 | 100 | 23-145 | 4 | 20 | |
| Carbon tetrachloride | mg/kg | ND | 0.047 | 0.049 | 0.033 | 0.031 | 70 | 63 | 43-148 | 5 | 20 | |
| Chlorobenzene | mg/kg | ND | 0.047 | 0.049 | 0.035 | 0.035 | 76 | 71 | 28-136 | 1 | 20 | |
| Chloroethane | mg/kg | ND | 0.047 | 0.049 | 0.064 | 0.071 | 138 | 144 | 34-160 | 11 | 20 | |
| Chloroform | mg/kg | ND | 0.047 | 0.049 | 0.042 | 0.042 | 90 | 84 | 54-136 | 1 | 20 | |
| Chloromethane | mg/kg | ND | 0.047 | 0.049 | 0.046 | 0.044 | 98 | 89 | 36-145 | 3 | 20 | |
| cis-1,2-Dichloroethene | mg/kg | ND | 0.047 | 0.049 | 0.042 | 0.041 | 89 | 83 | 52-132 | 1 | 20 | |
| cis-1,3-Dichloropropene | mg/kg | ND | 0.047 | 0.049 | 0.023 | 0.023 | 49 | 47 | 22-163 | 1 | 20 | |
| Dibromochloromethane | mg/kg | ND | 0.047 | 0.049 | 0.029 | 0.030 | 62 | 61 | 18-161 | 6 | 20 | |
| Dibromomethane | mg/kg | ND | 0.047 | 0.049 | 0.040 | 0.039 | 85 | 79 | 32-147 | 1 | 20 | |
| Dichlorodifluoromethane | mg/kg | ND | 0.047 | 0.049 | 0.015 | 0.015 | 32 | 30 | 10-138 | 1 | 20 | |
| Ethyl methacrylate | mg/kg | ND | 0.047 | 0.049 | <0.093 | <0.099 | 0 | 0 | 10-167 | | 20 | M1 |
| Ethylbenzene | mg/kg | ND | 0.047 | 0.049 | 0.035 | 0.034 | 76 | 69 | 24-150 | 3 | 20 | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

## REPORT OF LABORATORY ANALYSIS

**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3585893 3585894

| Parameter | Units | 50369748001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Iodomethane | mg/kg | ND | 0.047 | 0.049 | <0.093 | <0.099 | 78 | 72 | 23-142 | | 20 | |
| Methyl methacrylate | mg/kg | ND | 0.047 | 0.049 | <0.093 | <0.099 | 25 | 19 | 10-177 | | 20 | |
| Methyl-tert-butyl ether | mg/kg | ND | 0.047 | 0.049 | 0.037 | 0.051 | 79 | 102 | 57-141 | 31 | 20 | R1 |
| Methylene Chloride | mg/kg | ND | 0.047 | 0.049 | 0.045 | 0.063 | 96 | 127 | 40-140 | 34 | 20 | R1 |
| Styrene | mg/kg | ND | 0.047 | 0.049 | 0.032 | 0.032 | 69 | 64 | 12-138 | 1 | 20 | |
| Tetrachloroethene | mg/kg | ND | 0.047 | 0.049 | 0.039 | 0.038 | 84 | 77 | 26-159 | 3 | 20 | |
| Toluene | mg/kg | ND | 0.047 | 0.049 | 0.039 | 0.039 | 83 | 77 | 28-150 | 1 | 20 | |
| trans-1,2-Dichloroethene | mg/kg | ND | 0.047 | 0.049 | 0.042 | 0.056 | 89 | 113 | 50-134 | 30 | 20 | R1 |
| trans-1,3-Dichloropropene | mg/kg | ND | 0.047 | 0.049 | 0.023 | 0.024 | 50 | 48 | 17-153 | 3 | 20 | |
| trans-1,4-Dichloro-2-butene | mg/kg | ND | 0.047 | 0.049 | <0.093 | <0.099 | 21 | 18 | 10-146 | | 20 | |
| Trichloroethene | mg/kg | ND | 0.047 | 0.049 | 0.039 | 0.039 | 83 | 79 | 33-155 | 1 | 20 | |
| Trichlorofluoromethane | mg/kg | ND | 0.047 | 0.049 | 0.050 | 0.053 | 108 | 107 | 37-163 | 6 | 20 | |
| Vinyl acetate | mg/kg | ND | 0.19 | 0.2 | <0.093 | <0.099 | 16 | 15 | 10-183 | | 20 | |
| Vinyl chloride | mg/kg | ND | 0.047 | 0.049 | 0.054 | 0.052 | 115 | 106 | 37-161 | 2 | 20 | |
| Xylene (Total) | mg/kg | ND | 0.14 | 0.15 | 0.10 | 0.098 | 71 | 66 | 25-142 | 1 | 20 | |
| 4-Bromofluorobenzene (S) | %. | | | | | | 93 | 93 | 63-132 | | | |
| Dibromofluoromethane (S) | %. | | | | | | 98 | 98 | 75-135 | | | |
| Toluene-d8 (S) | %. | | | | | | 97 | 96 | 65-148 | | | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

## REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 784084 | Analysis Method: EPA 5030/8260
QC Batch Method: EPA 5030/8260 | Analysis Description: 8260 MSV TCLP
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

METHOD BLANK: 3587226 Matrix: Water
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| 1,1-Dichloroethene | mg/L | <0.050 | 0.050 | 04/09/24 11:01 | |
| 1,2-Dichloroethane | mg/L | <0.050 | 0.050 | 04/09/24 11:01 | |
| 2-Butanone (MEK) | mg/L | <1.0 | 1.0 | 04/09/24 11:01 | |
| Benzene | mg/L | <0.050 | 0.050 | 04/09/24 11:01 | |
| Carbon tetrachloride | mg/L | <0.050 | 0.050 | 04/09/24 11:01 | |
| Chlorobenzene | mg/L | <0.050 | 0.050 | 04/09/24 11:01 | |
| Chloroform | mg/L | <0.050 | 0.050 | 04/09/24 11:01 | |
| Tetrachloroethene | mg/L | <0.050 | 0.050 | 04/09/24 11:01 | |
| Trichloroethene | mg/L | <0.050 | 0.050 | 04/09/24 11:01 | |
| Vinyl chloride | mg/L | <0.020 | 0.020 | 04/09/24 11:01 | |
| 4-Bromofluorobenzene (S) | %. | 99 | 79-124 | 04/09/24 11:01 | |
| Dibromofluoromethane (S) | %. | 102 | 82-128 | 04/09/24 11:01 | |
| Toluene-d8 (S) | %. | 96 | 73-122 | 04/09/24 11:01 | |

LABORATORY CONTROL SAMPLE: 3587227

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| 1,1-Dichloroethene | mg/L | 0.5 | 0.56 | 113 | 71-130 | |
| 1,2-Dichloroethane | mg/L | 0.5 | 0.55 | 111 | 72-123 | |
| 2-Butanone (MEK) | mg/L | 2.5 | 2.3 | 90 | 67-135 | |
| Benzene | mg/L | 0.5 | 0.57 | 114 | 76-122 | |
| Carbon tetrachloride | mg/L | 0.5 | 0.56 | 113 | 73-127 | |
| Chlorobenzene | mg/L | 0.5 | 0.53 | 107 | 76-118 | |
| Chloroform | mg/L | 0.5 | 0.56 | 111 | 78-121 | |
| Tetrachloroethene | mg/L | 0.5 | 0.55 | 110 | 71-122 | |
| Trichloroethene | mg/L | 0.5 | 0.56 | 111 | 74-125 | |
| Vinyl chloride | mg/L | 0.5 | 0.54 | 109 | 55-139 | |
| 4-Bromofluorobenzene (S) | %. | | | 102 | 79-124 | |
| Dibromofluoromethane (S) | %. | | | 97 | 82-128 | |
| Toluene-d8 (S) | %. | | | 99 | 73-122 | |

MATRIX SPIKE SAMPLE: 3587228

| Parameter | Units | 50369577001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| 1,1-Dichloroethene | mg/L | ND | 0.5 | 0.54 | 107 | 53-144 | |
| 1,2-Dichloroethane | mg/L | ND | 0.5 | 0.54 | 108 | 50-138 | |
| 2-Butanone (MEK) | mg/L | ND | 2.5 | 2.0 | 81 | 45-138 | |

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## REPORT OF LABORATORY ANALYSIS

**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

MATRIX SPIKE SAMPLE: 3587228

| Parameter | Units | 50369577001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Benzene | mg/L | ND | 0.5 | 0.53 | 107 | 53-138 | |
| Carbon tetrachloride | mg/L | ND | 0.5 | 0.53 | 105 | 43-148 | |
| Chlorobenzene | mg/L | ND | 0.5 | 0.51 | 103 | 52-131 | |
| Chloroform | mg/L | ND | 0.5 | 0.55 | 110 | 54-138 | |
| Tetrachloroethene | mg/L | ND | 0.5 | 0.52 | 104 | 44-138 | |
| Trichloroethene | mg/L | ND | 0.5 | 0.54 | 107 | 49-140 | |
| Vinyl chloride | mg/L | ND | 0.5 | 0.54 | 108 | 41-147 | |
| 4-Bromofluorobenzene (S) | %. | | | | 99 | 79-124 | |
| Dibromofluoromethane (S) | %. | | | | 100 | 82-128 | |
| Toluene-d8 (S) | %. | | | | 98 | 73-122 | |

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**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 783413
QC Batch Method: EPA 3546
Analysis Method: EPA 8081
Analysis Description: 8081 GCS Pesticides
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50369587001

METHOD BLANK: 3584390 Matrix: Solid
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| 4,4'-DDD | ug/kg | <5.0 | 5.0 | 04/04/24 20:51 | |
| 4,4'-DDE | ug/kg | <5.0 | 5.0 | 04/04/24 20:51 | |
| 4,4'-DDT | ug/kg | <5.0 | 5.0 | 04/04/24 20:51 | |
| Aldrin | ug/kg | <2.5 | 2.5 | 04/04/24 20:51 | |
| alpha-BHC | ug/kg | <2.5 | 2.5 | 04/04/24 20:51 | |
| alpha-Chlordane | ug/kg | <2.5 | 2.5 | 04/04/24 20:51 | |
| beta-BHC | ug/kg | <2.5 | 2.5 | 04/04/24 20:51 | |
| Chlordane (Technical) | ug/kg | <50.0 | 50.0 | 04/04/24 20:51 | |
| delta-BHC | ug/kg | <2.5 | 2.5 | 04/04/24 20:51 | |
| Dieldrin | ug/kg | <5.0 | 5.0 | 04/04/24 20:51 | |
| Endosulfan I | ug/kg | <2.5 | 2.5 | 04/04/24 20:51 | |
| Endosulfan II | ug/kg | <5.0 | 5.0 | 04/04/24 20:51 | |
| Endosulfan sulfate | ug/kg | <5.0 | 5.0 | 04/04/24 20:51 | |
| Endrin | ug/kg | <5.0 | 5.0 | 04/04/24 20:51 | |
| Endrin aldehyde | ug/kg | <5.0 | 5.0 | 04/04/24 20:51 | |
| gamma-BHC (Lindane) | ug/kg | <2.5 | 2.5 | 04/04/24 20:51 | |
| gamma-Chlordane | ug/kg | <2.5 | 2.5 | 04/04/24 20:51 | |
| Heptachlor | ug/kg | <2.5 | 2.5 | 04/04/24 20:51 | |
| Heptachlor epoxide | ug/kg | <2.5 | 2.5 | 04/04/24 20:51 | |
| Toxaphene | ug/kg | <50.0 | 50.0 | 04/04/24 20:51 | |
| Decachlorobiphenyl (S) | %. | 70 | 10-166 | 04/04/24 20:51 | |

LABORATORY CONTROL SAMPLE: 3584391

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| 4,4'-DDD | ug/kg | 20 | 20.4 | 102 | 45-171 | |
| 4,4'-DDE | ug/kg | 20 | 20.7 | 104 | 45-163 | |
| 4,4'-DDT | ug/kg | 20 | 26.3 | 131 | 33-183 | |
| Aldrin | ug/kg | 10 | 9.9 | 99 | 50-151 | |
| alpha-BHC | ug/kg | 10 | 9.5 | 95 | 49-148 | |
| alpha-Chlordane | ug/kg | 10 | 10.1 | 101 | 46-150 | |
| beta-BHC | ug/kg | 10 | 10.9 | 109 | 51-158 | |
| delta-BHC | ug/kg | 10 | 10.0 | 100 | 25-154 | |
| Dieldrin | ug/kg | 20 | 21.1 | 106 | 48-164 | |
| Endosulfan I | ug/kg | 10 | 10.3 | 103 | 45-160 | |
| Endosulfan II | ug/kg | 20 | 21.4 | 107 | 49-163 | |
| Endosulfan sulfate | ug/kg | 20 | 21.8 | 109 | 41-154 | |
| Endrin | ug/kg | 20 | 21.7 | 109 | 48-163 | |
| Endrin aldehyde | ug/kg | 20 | 23.0 | 115 | 32-180 | |

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## REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

LABORATORY CONTROL SAMPLE: 3584391

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| gamma-BHC (Lindane) | ug/kg | 10 | 10.7 | 107 | 51-152 | |
| gamma-Chlordane | ug/kg | 10 | 10.5 | 105 | 44-173 | |
| Heptachlor | ug/kg | 10 | 11.3 | 113 | 52-155 | |
| Heptachlor epoxide | ug/kg | 10 | 10.4 | 104 | 47-154 | |
| Decachlorobiphenyl (S) | %. | | | 72 | 10-166 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3584392 3584393

| Parameter | Units | 50369747001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4,4'-DDD | ug/kg | ND | 78.4 | 69 | 65.6 | 65.9 | 84 | 95 | 10-145 | 0 | 20 | |
| 4,4'-DDE | ug/kg | ND | 78.4 | 69 | 66.1 | 66.3 | 84 | 96 | 10-136 | 0 | 20 | |
| 4,4'-DDT | ug/kg | ND | 78.4 | 69 | 82.6 | 81.9 | 105 | 119 | 10-160 | 1 | 20 | |
| Aldrin | ug/kg | ND | 39.2 | 34.5 | 37.6 | 35.8 | 96 | 104 | 10-155 | 5 | 20 | |
| alpha-BHC | ug/kg | ND | 39.2 | 34.5 | 33.2 | 32.5 | 85 | 94 | 10-151 | 2 | 20 | |
| alpha-Chlordane | ug/kg | ND | 39.2 | 34.5 | 33.6 | 32.9 | 86 | 95 | 10-148 | 2 | 20 | |
| beta-BHC | ug/kg | ND | 39.2 | 34.5 | 36.3 | 34.3 | 92 | 99 | 10-160 | 6 | 20 | |
| delta-BHC | ug/kg | ND | 39.2 | 34.5 | 32.4 | 32.8 | 83 | 95 | 10-122 | 1 | 20 | |
| Dieldrin | ug/kg | ND | 78.4 | 69 | 68.7 | 67.8 | 88 | 98 | 10-182 | 1 | 20 | |
| Endosulfan I | ug/kg | ND | 39.2 | 34.5 | 33.0 | 32.5 | 84 | 94 | 10-145 | 2 | 20 | |
| Endosulfan II | ug/kg | ND | 78.4 | 69 | 67.8 | 67.7 | 86 | 98 | 10-168 | 0 | 20 | |
| Endosulfan sulfate | ug/kg | ND | 78.4 | 69 | 63.7 | 64.4 | 81 | 93 | 13-124 | 1 | 20 | |
| Endrin | ug/kg | ND | 78.4 | 69 | 69.6 | 68.8 | 89 | 100 | 10-148 | 1 | 20 | |
| Endrin aldehyde | ug/kg | ND | 78.4 | 69 | 71.9 | 70.7 | 92 | 102 | 10-169 | 2 | 20 | |
| gamma-BHC (Lindane) | ug/kg | ND | 39.2 | 34.5 | 36.6 | 35.5 | 93 | 103 | 10-153 | 3 | 20 | |
| gamma-Chlordane | ug/kg | ND | 39.2 | 34.5 | 35.7 | 34.0 | 91 | 98 | 10-173 | 5 | 20 | |
| Heptachlor | ug/kg | ND | 39.2 | 34.5 | 38.0 | 37.1 | 97 | 108 | 10-158 | 2 | 20 | |
| Heptachlor epoxide | ug/kg | ND | 39.2 | 34.5 | 34.1 | 33.6 | 87 | 98 | 10-172 | 1 | 20 | |
| Decachlorobiphenyl (S) | %. | | | | | | 51 | 56 | 10-166 | | | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3584394 3584395

| Parameter | Units | 50369748001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4,4'-DDD | ug/kg | ND | 80 | 71.4 | 61.4 | 60.2 | 77 | 84 | 10-145 | 2 | 20 | |
| 4,4'-DDE | ug/kg | ND | 80 | 71.4 | 62.1 | 60.9 | 78 | 85 | 10-136 | 2 | 20 | |
| 4,4'-DDT | ug/kg | ND | 80 | 71.4 | 71.6 | 68.7 | 89 | 96 | 10-160 | 4 | 20 | |
| Aldrin | ug/kg | ND | 40 | 35.7 | 32.7 | 31.2 | 82 | 87 | 10-155 | 5 | 20 | |
| alpha-BHC | ug/kg | ND | 40 | 35.7 | 30.4 | 29.9 | 76 | 84 | 10-151 | 2 | 20 | |
| alpha-Chlordane | ug/kg | ND | 40 | 35.7 | 31.3 | 30.7 | 78 | 86 | 10-148 | 2 | 20 | |
| beta-BHC | ug/kg | ND | 40 | 35.7 | 32.1 | 31.3 | 80 | 88 | 10-160 | 3 | 20 | |
| delta-BHC | ug/kg | ND | 40 | 35.7 | 29.3 | 28.7 | 73 | 80 | 10-122 | 2 | 20 | |
| Dieldrin | ug/kg | ND | 80 | 71.4 | 62.9 | 61.7 | 79 | 86 | 10-182 | 2 | 20 | |
| Endosulfan I | ug/kg | ND | 40 | 35.7 | 30.3 | 29.5 | 76 | 83 | 10-145 | 3 | 20 | |

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3584394 3584395

| Parameter | Units | 50369748001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Endosulfan II | ug/kg | ND | 80 | 71.4 | 63.9 | 61.1 | 80 | 86 | 10-168 | 4 | 20 | |
| Endosulfan sulfate | ug/kg | ND | 80 | 71.4 | 57.9 | 54.8 | 72 | 77 | 13-124 | 6 | 20 | |
| Endrin | ug/kg | ND | 80 | 71.4 | 64.0 | 62.9 | 80 | 88 | 10-148 | 2 | 20 | |
| Endrin aldehyde | ug/kg | ND | 80 | 71.4 | 67.3 | 63.4 | 84 | 89 | 10-169 | 6 | 20 | |
| gamma-BHC (Lindane) | ug/kg | ND | 40 | 35.7 | 32.9 | 32.1 | 82 | 90 | 10-153 | 3 | 20 | |
| gamma-Chlordane | ug/kg | ND | 40 | 35.7 | 32.1 | 30.4 | 80 | 85 | 10-173 | 5 | 20 | |
| Heptachlor | ug/kg | ND | 40 | 35.7 | 35.5 | 35.3 | 89 | 99 | 10-158 | 1 | 20 | |
| Heptachlor epoxide | ug/kg | ND | 40 | 35.7 | 31.1 | 30.5 | 78 | 85 | 10-172 | 2 | 20 | |
| Decachlorobiphenyl (S) | %. | | | | | | 48 | 47 | 10-166 | | | |

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 784116
QC Batch Method: EPA 3510
Analysis Method: EPA 8081
Analysis Description: 8081 GCS TCLP Pesticides
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50369587001

METHOD BLANK: 3587350 Matrix: Water
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Chlordane (Technical) | mg/L | <0.0010 | 0.0010 | 04/10/24 11:32 | |
| Endrin | mg/L | <0.00010 | 0.00010 | 04/10/24 11:32 | |
| gamma-BHC (Lindane) | mg/L | <0.000050 | 0.000050 | 04/10/24 11:32 | |
| Heptachlor | mg/L | <0.000050 | 0.000050 | 04/10/24 11:32 | |
| Heptachlor epoxide | mg/L | <0.000050 | 0.000050 | 04/10/24 11:32 | |
| Methoxychlor | mg/L | <0.00050 | 0.00050 | 04/10/24 11:32 | |
| Toxaphene | mg/L | <0.0010 | 0.0010 | 04/10/24 11:32 | |
| Decachlorobiphenyl (S) | %. | 42 | 10-124 | 04/10/24 11:32 | |

LABORATORY CONTROL SAMPLE: 3587351

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Endrin | mg/L | 0.002 | 0.0022 | 111 | 29-177 | |
| gamma-BHC (Lindane) | mg/L | 0.001 | 0.0011 | 107 | 26-163 | |
| Heptachlor | mg/L | 0.001 | 0.00075 | 75 | 12-157 | |
| Heptachlor epoxide | mg/L | 0.001 | 0.0011 | 106 | 27-168 | |
| Methoxychlor | mg/L | 0.01 | 0.012 | 120 | 28-174 | |
| Decachlorobiphenyl (S) | %. | | | 45 | 10-124 | |

MATRIX SPIKE SAMPLE: 3587352

| Parameter | Units | 50369587001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Endrin | mg/L | <0.00050 | 0.01 | 0.012 | 116 | 10-179 | |
| gamma-BHC (Lindane) | mg/L | <0.00025 | 0.005 | 0.0059 | 119 | 11-157 | |
| Heptachlor | mg/L | <0.00025 | 0.005 | 0.0045 | 91 | 10-154 | |
| Heptachlor epoxide | mg/L | <0.00025 | 0.005 | 0.0054 | 109 | 10-170 | |
| Methoxychlor | mg/L | <0.0025 | 0.05 | 0.066 | 132 | 10-162 | |
| Decachlorobiphenyl (S) | %. | | | | 42 | 10-124 | |

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**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 784646
QC Batch Method: EPA 3546
Analysis Method: EPA 8082
Analysis Description: 8082 PCB Solids
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

METHOD BLANK: 3589719 Matrix: Solid

Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| PCB-1016 (Aroclor 1016) | ug/kg | <100 | 100 | 04/12/24 22:16 | |
| PCB-1221 (Aroclor 1221) | ug/kg | <100 | 100 | 04/12/24 22:16 | |
| PCB-1232 (Aroclor 1232) | ug/kg | <100 | 100 | 04/12/24 22:16 | |
| PCB-1242 (Aroclor 1242) | ug/kg | <100 | 100 | 04/12/24 22:16 | |
| PCB-1248 (Aroclor 1248) | ug/kg | <100 | 100 | 04/12/24 22:16 | |
| PCB-1254 (Aroclor 1254) | ug/kg | <100 | 100 | 04/12/24 22:16 | |
| PCB-1260 (Aroclor 1260) | ug/kg | <100 | 100 | 04/12/24 22:16 | |
| Tetrachloro-m-xylene (S) | %. | 89 | 10-133 | 04/12/24 22:16 | |

LABORATORY CONTROL SAMPLE: 3589720

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| PCB-1016 (Aroclor 1016) | ug/kg | 333 | 318 | 95 | 50-120 | |
| PCB-1260 (Aroclor 1260) | ug/kg | 333 | 379 | 114 | 40-122 | |
| Tetrachloro-m-xylene (S) | %. | | | 89 | 10-133 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3589721 3589722

| Parameter | Units | 50369585001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PCB-1016 (Aroclor 1016) | ug/kg | ND | 373 | 366 | 352 | 322 | 95 | 88 | 10-154 | 9 | 20 | |
| PCB-1260 (Aroclor 1260) | ug/kg | ND | 373 | 366 | 342 | 322 | 92 | 88 | 10-165 | 6 | 20 | |
| Tetrachloro-m-xylene (S) | %. | | | | | | 90 | 87 | 10-133 | | | |

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 784552
QC Batch Method: EPA 8151
Analysis Method: EPA 8151A
Analysis Description: 8151 GCS TCLP Herbicides
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50369587001

METHOD BLANK: 3589141 Matrix: Water
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| 2,4,5-TP (Silvex) | mg/L | <0.0010 | 0.0010 | 04/13/24 16:35 | |
| 2,4-D | mg/L | <0.0010 | 0.0010 | 04/13/24 16:35 | |
| 2,4-DCAA (S) | %. | 77 | 22-132 | 04/13/24 16:35 | |

LABORATORY CONTROL SAMPLE: 3589142

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| 2,4,5-TP (Silvex) | mg/L | 0.005 | 0.0049 | 98 | 37-122 | |
| 2,4-D | mg/L | 0.005 | 0.0048 | 96 | 24-129 | |
| 2,4-DCAA (S) | %. | | | 75 | 22-132 | |

MATRIX SPIKE SAMPLE: 3589143

| Parameter | Units | 50369581001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| 2,4,5-TP (Silvex) | mg/L | ND | 0.025 | 0.013 | 54 | 20-155 | |
| 2,4-D | mg/L | 0.14 mg/kg | 0.025 | 0.020 | 82 | 12-159 | |
| 2,4-DCAA (S) | %. | | | | 27 | 22-132 | |

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## REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 783934
QC Batch Method: EPA 3546
Analysis Method: EPA 8270
Analysis Description: 8270 Solid MSSV Microwave Short Spike
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

METHOD BLANK: 3586852
Matrix: Solid

Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| 1-Methylnaphthalene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2,4,5-Trichlorophenol | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2,4,6-Trichlorophenol | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2,4-Dichlorophenol | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2,4-Dimethylphenol | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2,4-Dinitrophenol | ug/kg | <1600 | 1600 | 04/10/24 16:41 | |
| 2,4-Dinitrotoluene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2,6-Dinitrotoluene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2-Chloronaphthalene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2-Chlorophenol | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2-Methylnaphthalene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2-Methylphenol(o-Cresol) | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2-Nitroaniline | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2-Nitrophenol | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 3&4-Methylphenol(m&p Cresol) | ug/kg | <660 | 660 | 04/10/24 16:41 | |
| 3,3'-Dichlorobenzidine | ug/kg | <660 | 660 | 04/10/24 16:41 | |
| 3-Nitroaniline | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 4,6-Dinitro-2-methylphenol | ug/kg | <660 | 660 | 04/10/24 16:41 | |
| 4-Bromophenylphenyl ether | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 4-Chloro-3-methylphenol | ug/kg | <660 | 660 | 04/10/24 16:41 | |
| 4-Chloroaniline | ug/kg | <660 | 660 | 04/10/24 16:41 | |
| 4-Chlorophenylphenyl ether | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 4-Nitroaniline | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 4-Nitrophenol | ug/kg | <1600 | 1600 | 04/10/24 16:41 | |
| Acenaphthene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Acenaphthylene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Anthracene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Benzo(a)anthracene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Benzo(a)pyrene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Benzo(b)fluoranthene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Benzo(g,h,i)perylene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Benzo(k)fluoranthene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Benzyl alcohol | ug/kg | <660 | 660 | 04/10/24 16:41 | |
| bis(2-Chloroethoxy)methane | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| bis(2-Chloroethyl) ether | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| bis(2-Ethylhexyl)phthalate | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| bis(2chloro1methylethyl) ether | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Butylbenzylphthalate | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Chrysene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Di-n-butylphthalate | ug/kg | <330 | 330 | 04/10/24 16:41 | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

## REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

METHOD BLANK: 3586852 Matrix: Solid
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Di-n-octylphthalate | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Dibenz(a,h)anthracene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Dibenzofuran | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Diethylphthalate | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Dimethylphthalate | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Fluoranthene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Fluorene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Hexachloro-1,3-butadiene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Hexachlorobenzene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Hexachlorocyclopentadiene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Hexachloroethane | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Indeno(1,2,3-cd)pyrene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Isophorone | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| N-Nitroso-di-n-propylamine | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| N-Nitrosodiphenylamine | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Naphthalene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Nitrobenzene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Pentachlorophenol | ug/kg | <1600 | 1600 | 04/10/24 16:41 | |
| Phenanthrene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Phenol | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| Pyrene | ug/kg | <330 | 330 | 04/10/24 16:41 | |
| 2,4,6-Tribromophenol (S) | %. | 82 | 10-121 | 04/10/24 16:41 | |
| 2-Fluorobiphenyl (S) | %. | 80 | 33-102 | 04/10/24 16:41 | |
| 2-Fluorophenol (S) | %. | 81 | 21-109 | 04/10/24 16:41 | |
| Nitrobenzene-d5 (S) | %. | 79 | 32-95 | 04/10/24 16:41 | |
| p-Terphenyl-d14 (S) | %. | 94 | 20-120 | 04/10/24 16:41 | |
| Phenol-d5 (S) | %. | 88 | 27-116 | 04/10/24 16:41 | |

LABORATORY CONTROL SAMPLE: 3586853

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| 2,4-Dinitrotoluene | ug/kg | 1670 | 1370 | 82 | 59-121 | |
| 2-Chlorophenol | ug/kg | 1670 | 1340 | 80 | 57-102 | |
| 2-Methylnaphthalene | ug/kg | 1670 | 1340 | 81 | 52-118 | |
| 4-Chloro-3-methylphenol | ug/kg | 1670 | 1500 | 90 | 58-135 | |
| 4-Nitrophenol | ug/kg | 1670 | <1600 | 63 | 49-133 | |
| Acenaphthene | ug/kg | 1670 | 1440 | 86 | 63-105 | |
| Acenaphthylene | ug/kg | 1670 | 1360 | 81 | 61-104 | |
| Anthracene | ug/kg | 1670 | 1450 | 87 | 66-104 | |
| Benzo(a)anthracene | ug/kg | 1670 | 1460 | 87 | 69-109 | |
| Benzo(a)pyrene | ug/kg | 1670 | 1410 | 85 | 63-105 | |
| Benzo(b)fluoranthene | ug/kg | 1670 | 1550 | 93 | 63-115 | |
| Benzo(g,h,i)perylene | ug/kg | 1670 | 1480 | 89 | 64-112 | |
| Benzo(k)fluoranthene | ug/kg | 1670 | 1400 | 84 | 65-115 | |

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Grand Rapids, MI 49512
(616)975-4500

## QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

LABORATORY CONTROL SAMPLE: 3586853

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Chrysene | ug/kg | 1670 | 1400 | 84 | 68-109 | |
| Dibenz(a,h)anthracene | ug/kg | 1670 | 1490 | 89 | 65-112 | |
| Fluoranthene | ug/kg | 1670 | 1540 | 92 | 67-116 | |
| Fluorene | ug/kg | 1670 | 1480 | 89 | 66-112 | |
| Indeno(1,2,3-cd)pyrene | ug/kg | 1670 | 1460 | 88 | 66-113 | |
| N-Nitroso-di-n-propylamine | ug/kg | 1670 | 1260 | 76 | 45-112 | |
| Naphthalene | ug/kg | 1670 | 1280 | 77 | 58-103 | |
| Pentachlorophenol | ug/kg | 1670 | <1600 | 72 | 31-123 | |
| Phenanthrene | ug/kg | 1670 | 1470 | 88 | 68-106 | |
| Phenol | ug/kg | 1670 | 1410 | 84 | 49-109 | |
| Pyrene | ug/kg | 1670 | 1420 | 85 | 67-110 | |
| 2,4,6-Tribromophenol (S) | %. | | | 89 | 10-121 | |
| 2-Fluorobiphenyl (S) | %. | | | 85 | 33-102 | |
| 2-Fluorophenol (S) | %. | | | 87 | 21-109 | |
| Nitrobenzene-d5 (S) | %. | | | 83 | 32-95 | |
| p-Terphenyl-d14 (S) | %. | | | 97 | 20-120 | |
| Phenol-d5 (S) | %. | | | 94 | 27-116 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3586854 3586855

| Parameter | Units | 50369349001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,4-Dinitrotoluene | ug/kg | ND | 1870 | 1890 | 1220 | 1410 | 66 | 74 | 10-132 | 14 | 20 | |
| 2-Chlorophenol | ug/kg | ND | 1860 | 1890 | 510 | 556 | 27 | 29 | 11-121 | 9 | 20 | |
| 2-Methylnaphthalene | ug/kg | ND | 1860 | 1890 | 1440 | 1490 | 77 | 78 | 18-130 | 3 | 20 | |
| 4-Chloro-3-methylphenol | ug/kg | ND | 1860 | 1890 | 1700 | 1720 | 91 | 91 | 17-143 | 1 | 20 | |
| 4-Nitrophenol | ug/kg | ND | 1860 | 1890 | <1790 | <1820 | 8 | 10 | 10-148 | | 20 | M1 |
| Acenaphthene | ug/kg | ND | 1860 | 1890 | 1340 | 1490 | 72 | 78 | 12-129 | 10 | 20 | |
| Acenaphthylene | ug/kg | ND | 1860 | 1890 | 1290 | 1450 | 69 | 77 | 10-126 | 11 | 20 | |
| Anthracene | ug/kg | ND | 1860 | 1890 | 1630 | 1720 | 76 | 79 | 10-137 | 6 | 20 | |
| Benzo(a)anthracene | ug/kg | 0.58 mg/kg | 1860 | 1890 | 1770 | 1950 | 63 | 72 | 10-140 | 10 | 20 | |
| Benzo(a)pyrene | ug/kg | 0.54 mg/kg | 1860 | 1890 | 1750 | 1920 | 65 | 73 | 10-137 | 9 | 20 | |
| Benzo(b)fluoranthene | ug/kg | 0.77 mg/kg | 1860 | 1890 | 2150 | 2220 | 74 | 76 | 10-143 | 3 | 20 | |
| Benzo(g,h,i)perylene | ug/kg | ND | 1860 | 1890 | 1710 | 1880 | 72 | 79 | 10-143 | 9 | 20 | |
| Benzo(k)fluoranthene | ug/kg | ND | 1860 | 1890 | 1540 | 1730 | 67 | 76 | 10-146 | 12 | 20 | |
| Chrysene | ug/kg | 0.62 mg/kg | 1860 | 1890 | 1860 | 2020 | 66 | 74 | 10-143 | 9 | 20 | |
| Dibenz(a,h)anthracene | ug/kg | ND | 1860 | 1890 | 1550 | 1640 | 83 | 87 | 13-125 | 6 | 20 | |
| Fluoranthene | ug/kg | 1.4 mg/kg | 1860 | 1890 | 2430 | 2590 | 58 | 66 | 10-151 | 7 | 20 | |
| Fluorene | ug/kg | ND | 1860 | 1890 | 1490 | 1610 | 80 | 85 | 10-136 | 8 | 20 | |
| Indeno(1,2,3-cd)pyrene | ug/kg | ND | 1860 | 1890 | 1680 | 1780 | 74 | 78 | 10-139 | 6 | 20 | |
| N-Nitroso-di-n-propylamine | ug/kg | ND | 1860 | 1890 | 1270 | 1410 | 68 | 75 | 13-125 | 11 | 20 | |
| Naphthalene | ug/kg | ND | 1860 | 1890 | 1420 | 1470 | 76 | 78 | 17-123 | 3 | 20 | |

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3586854 3586855

| Parameter | Units | 50369349001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pentachlorophenol | ug/kg | ND | 1860 | 1890 | <1790 | <1820 | 6 | 4 | 10-141 | | 20 | M1 |
| Phenanthrene | ug/kg | 1.1 mg/kg | 1860 | 1890 | 2350 | 2490 | 68 | 75 | 10-149 | 6 | 20 | |
| Phenol | ug/kg | ND | 1860 | 1890 | 1150 | 1240 | 62 | 65 | 16-120 | 8 | 20 | |
| Pyrene | ug/kg | 1.1 mg/kg | 1860 | 1890 | 2160 | 2420 | 59 | 71 | 10-152 | 11 | 20 | |
| 2,4,6-Tribromophenol (S) | %. | | | | | | 6 | 6 | 10-121 | | | S2 |
| 2-Fluorobiphenyl (S) | %. | | | | | | 65 | 72 | 33-102 | | | |
| 2-Fluorophenol (S) | %. | | | | | | 21 | 22 | 21-109 | | | |
| Nitrobenzene-d5 (S) | %. | | | | | | 72 | 78 | 32-95 | | | |
| p-Terphenyl-d14 (S) | %. | | | | | | 82 | 84 | 20-120 | | | |
| Phenol-d5 (S) | %. | | | | | | 67 | 71 | 27-116 | | | |

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

| | | | |
|---|---|---|---|
| QC Batch: | 784240 | Analysis Method: | EPA 8270 |
| QC Batch Method: | EPA 3510 | Analysis Description: | 8270 TCLP MSSV |
| | | Laboratory: | Pace Analytical Services - Indianapolis |

Associated Lab Samples: 50369587001

METHOD BLANK: 3587823 Matrix: Water

Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| 1,4-Dichlorobenzene | mg/L | <0.10 | 0.10 | 04/11/24 17:08 | |
| 2,4,5-Trichlorophenol | mg/L | <0.50 | 0.50 | 04/11/24 17:08 | |
| 2,4,6-Trichlorophenol | mg/L | <0.10 | 0.10 | 04/11/24 17:08 | |
| 2,4-Dinitrotoluene | mg/L | <0.10 | 0.10 | 04/11/24 17:08 | |
| 2-Methylphenol(o-Cresol) | mg/L | <0.10 | 0.10 | 04/11/24 17:08 | |
| 3&4-Methylphenol(m&p Cresol) | mg/L | <0.20 | 0.20 | 04/11/24 17:08 | |
| Hexachloro-1,3-butadiene | mg/L | <0.10 | 0.10 | 04/11/24 17:08 | |
| Hexachlorobenzene | mg/L | <0.10 | 0.10 | 04/11/24 17:08 | |
| Hexachloroethane | mg/L | <0.10 | 0.10 | 04/11/24 17:08 | |
| Nitrobenzene | mg/L | <0.10 | 0.10 | 04/11/24 17:08 | |
| Pentachlorophenol | mg/L | <0.50 | 0.50 | 04/11/24 17:08 | |
| Pyridine | mg/L | <0.10 | 0.10 | 04/11/24 17:08 | |
| 2,4,6-Tribromophenol (S) | %. | 91 | 37-127 | 04/11/24 17:08 | |
| 2-Fluorobiphenyl (S) | %. | 63 | 36-92 | 04/11/24 17:08 | |
| 2-Fluorophenol (S) | %. | 50 | 23-69 | 04/11/24 17:08 | |
| Nitrobenzene-d5 (S) | %. | 85 | 44-107 | 04/11/24 17:08 | |
| p-Terphenyl-d14 (S) | %. | 100 | 57-146 | 04/11/24 17:08 | |
| Phenol-d5 (S) | %. | 34 | 15-47 | 04/11/24 17:08 | |

LABORATORY CONTROL SAMPLE: 3587824

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| 1,4-Dichlorobenzene | mg/L | 0.05 | 0.021 | 41 | 23-87 | |
| 2,4,5-Trichlorophenol | mg/L | 0.05 | <0.050 | 80 | 43-131 | |
| 2,4,6-Trichlorophenol | mg/L | 0.05 | 0.035 | 71 | 42-128 | |
| 2,4-Dinitrotoluene | mg/L | 0.05 | 0.035 | 70 | 47-116 | |
| 2-Methylphenol(o-Cresol) | mg/L | 0.05 | 0.029 | 59 | 35-98 | |
| 3&4-Methylphenol(m&p Cresol) | mg/L | 0.1 | 0.054 | 54 | 30-91 | |
| Hexachloro-1,3-butadiene | mg/L | 0.05 | 0.019 | 38 | 10-92 | |
| Hexachlorobenzene | mg/L | 0.05 | 0.029 | 57 | 34-89 | |
| Hexachloroethane | mg/L | 0.05 | 0.018 | 36 | 14-84 | |
| Nitrobenzene | mg/L | 0.05 | 0.037 | 74 | 39-113 | |
| Pentachlorophenol | mg/L | 0.05 | <0.050 | 81 | 16-133 | |
| Pyridine | mg/L | 0.05 | 0.026 | 52 | 10-61 | |
| 2,4,6-Tribromophenol (S) | %. | | | 76 | 37-127 | |
| 2-Fluorobiphenyl (S) | %. | | | 51 | 36-92 | |
| 2-Fluorophenol (S) | %. | | | 42 | 23-69 | |
| Nitrobenzene-d5 (S) | %. | | | 70 | 44-107 | |
| p-Terphenyl-d14 (S) | %. | | | 88 | 57-146 | |

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

LABORATORY CONTROL SAMPLE: 3587824

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Phenol-d5 (S) | %. | | | 29 | 15-47 | |

MATRIX SPIKE SAMPLE: 3587825

| Parameter | Units | 50369577001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| 1,4-Dichlorobenzene | mg/L | ND | 0.5 | 0.20 | 41 | 18-91 | |
| 2,4,5-Trichlorophenol | mg/L | ND | 0.5 | <0.50 | 90 | 51-131 | |
| 2,4,6-Trichlorophenol | mg/L | ND | 0.5 | 0.41 | 82 | 50-124 | |
| 2,4-Dinitrotoluene | mg/L | ND | 0.5 | 0.43 | 85 | 41-118 | |
| 2-Methylphenol(o-Cresol) | mg/L | ND | 0.5 | 0.35 | 70 | 23-115 | |
| 3&4-Methylphenol(m&p Cresol) | mg/L | ND | 1 | 0.63 | 63 | 18-110 | |
| Hexachloro-1,3-butadiene | mg/L | ND | 0.5 | 0.16 | 31 | 10-95 | |
| Hexachlorobenzene | mg/L | ND | 0.5 | 0.28 | 57 | 13-107 | |
| Hexachloroethane | mg/L | ND | 0.5 | 0.16 | 32 | 10-88 | |
| Nitrobenzene | mg/L | ND | 0.5 | 0.40 | 80 | 36-113 | |
| Pentachlorophenol | mg/L | ND | 0.5 | <0.50 | 74 | 27-139 | |
| Pyridine | mg/L | ND | 0.5 | 0.25 | 50 | 11-66 | |
| 2,4,6-Tribromophenol (S) | %. | | | | 85 | 37-127 | |
| 2-Fluorobiphenyl (S) | %. | | | | 65 | 36-92 | |
| 2-Fluorophenol (S) | %. | | | | 47 | 23-69 | |
| Nitrobenzene-d5 (S) | %. | | | | 79 | 44-107 | |
| p-Terphenyl-d14 (S) | %. | | | | 90 | 57-146 | |
| Phenol-d5 (S) | %. | | | | 33 | 15-47 | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

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## QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

| | | | |
|---|---|---|---|
| QC Batch: | 783511 | Analysis Method: | SM 2540G |
| QC Batch Method: | SM 2540G | Analysis Description: | Dry Weight/Percent Moisture |
| | | Laboratory: | Pace Analytical Services - Indianapolis |

Associated Lab Samples: 50369587001

SAMPLE DUPLICATE: 3584810

| Parameter | Units | 50369587001 Result | Dup Result | RPD | Max RPD | Qualifiers |
|---|---|---|---|---|---|---|
| Percent Moisture | % | 98.4 | 98.4 | 0 | 10 | N2 |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

| | | | |
|---|---|---|---|
| QC Batch: | 783512 | Analysis Method: | SM 2540G |
| QC Batch Method: | SM 2540G | Analysis Description: | 2540G Total Solids |
| | | Laboratory: | Pace Analytical Services - Indianapolis |

Associated Lab Samples: 50369587001

SAMPLE DUPLICATE: 3584811

| Parameter | Units | 50369587001 Result | Dup Result | RPD | Max RPD | Qualifiers |
|---|---|---|---|---|---|---|
| Total Solids | % | 1.6 | 1.6 | 4 | 10 | N2 |

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 783513
QC Batch Method: SM 2540G
Analysis Method: SM 2540G
Analysis Description: 2540G Total Volatile Solids
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

SAMPLE DUPLICATE: 3584813

| Parameter | Units | 50369587001 Result | Dup Result | RPD | Max RPD | Qualifiers |
|---|---|---|---|---|---|---|
| Total Volatile Solids | % | 58.8 | 56.7 | 4 | 10 | N2 |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 783903
QC Batch Method: EPA 9038
Analysis Method: EPA 9038
Analysis Description: 9038 S2ulfate Soil
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50369587001

METHOD BLANK: 3586752 Matrix: Solid
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Sulfate | mg/kg | <100 | 100 | 04/08/24 13:44 | N2 |

LABORATORY CONTROL SAMPLE: 3586753

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Sulfate | mg/kg | 200 | 206 | 103 | 90-110 | N2 |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3586754 3586755

| Parameter | Units | 50369795001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Sulfate | mg/kg | <916 | 1830 | 1830 | 2770 | 2700 | 143 | 139 | 90-110 | 2 | 20 | M3,N2 |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

## REPORT OF LABORATORY ANALYSIS

**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 784239
QC Batch Method: EPA 9045
Analysis Method: EPA 9045
Analysis Description: 9045 pH
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

SAMPLE DUPLICATE: 3587821

| Parameter | Units | 50370014001 Result | Dup Result | RPD | Max RPD | Qualifiers |
|---|---|---|---|---|---|---|
| pH at 25 Degrees C | Std. Units | 4.6 | 4.7 | 0 | 2 | H3 |

SAMPLE DUPLICATE: 3587822

| Parameter | Units | 50370020001 Result | Dup Result | RPD | Max RPD | Qualifiers |
|---|---|---|---|---|---|---|
| pH at 25 Degrees C | Std. Units | 9.2 | 9.2 | 0 | 2 | H3 |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 783628
QC Batch Method: EPA 351.2
Analysis Method: EPA 351.2
Analysis Description: 351.2 TKN
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50369587001

METHOD BLANK: 3585442 Matrix: Solid
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Nitrogen, Kjeldahl, Total | mg/kg | <49.0 | 49.0 | 04/06/24 11:15 | N2 |

LABORATORY CONTROL SAMPLE: 3585443

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Nitrogen, Kjeldahl, Total | mg/kg | 490 | 460 | 94 | 90-110 | N2 |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3585581 3585582

| Parameter | Units | 50369773001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Nitrogen, Kjeldahl, Total | mg/kg | 77100 | 14700 | 15100 | 94400 | 92500 | 118 | 102 | 90-110 | 2 | 20 | M0,N2 |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

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# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 784583
QC Batch Method: EPA 353.2
Analysis Method: EPA 353.2
Analysis Description: 353.2 Nitrate + Nitrite
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50369587001

METHOD BLANK: 3589343 Matrix: Solid
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Nitrogen, Nitrate | mg/kg | <5.0 | 5.0 | 04/11/24 20:25 | N2 |
| Nitrogen, NO2 plus NO3 | mg/kg | <5.0 | 5.0 | 04/11/24 20:25 | N2 |

LABORATORY CONTROL SAMPLE: 3589344

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Nitrogen, Nitrate | mg/kg | 9.9 | 10.0 | 101 | 90-110 | N2 |
| Nitrogen, NO2 plus NO3 | mg/kg | 19.9 | 19.7 | 99 | 90-110 | N2 |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3589345 3589346

| Parameter | Units | 50369916001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Nitrogen, Nitrate | mg/kg | ND | 86 | 86 | 90.8 | 89.9 | 106 | 105 | 90-110 | 1 | 20 | N2 |
| Nitrogen, NO2 plus NO3 | mg/kg | ND | 172 | 172 | 175 | 174 | 102 | 101 | 90-110 | 1 | 20 | N2 |

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## QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 785030
QC Batch Method: SM 4500-NH3 B
Analysis Method: SM 4500-NH3 G
Analysis Description: 4500 Ammonia, Distilled
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

METHOD BLANK: 3591554 Matrix: Solid
Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Nitrogen, Ammonia | mg/kg | <4.9 | 4.9 | 04/16/24 13:42 | N2 |

LABORATORY CONTROL SAMPLE: 3591555

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Nitrogen, Ammonia | mg/kg | 19.7 | 20.3 | 103 | 90-110 | N2 |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3591556 3591557

| Parameter | Units | 50369587001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Nitrogen, Ammonia | mg/kg | 46500 | 11300 | 11600 | 57500 | 59100 | 98 | 109 | 90-110 | 3 | 20 | N2 |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

**REPORT OF LABORATORY ANALYSIS**

**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

# QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 784900
QC Batch Method: SM 4500-Cl-E
Analysis Method: SM 4500-Cl-E
Analysis Description: 4500 Chloride
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

METHOD BLANK: 3591038 Matrix: Solid

Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Chloride | mg/kg | <99.4 | 99.4 | 04/15/24 14:27 | N2 |

LABORATORY CONTROL SAMPLE: 3591039

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Chloride | mg/kg | 199 | 213 | 107 | 90-110 | N2 |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3591040 3591041

| Parameter | Units | 50369587001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Chloride | mg/kg | 10300 | 12100 | 12100 | 23600 | 23900 | 110 | 113 | 90-110 | 1 | 20 | M0,N2 |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

## REPORT OF LABORATORY ANALYSIS

**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

## QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50369587

QC Batch: 784844
QC Batch Method: SM 4500-CN-C
Analysis Method: SM 4500-CN-E
Analysis Description: 4500CNE Cyanide
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50369587001

METHOD BLANK: 3590918 Matrix: Solid

Associated Lab Samples: 50369587001

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Cyanide | mg/kg | <0.49 | 0.49 | 04/15/24 13:53 | N2 |

LABORATORY CONTROL SAMPLE: 3590919

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Cyanide | mg/kg | 4.9 | 4.8 | 99 | 90-110 | N2 |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3590920 3590921

| Parameter | Units | 50369587001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Max RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Cyanide | mg/kg | <29.6 | 299 | 299 | 287 | 305 | 97 | 103 | 90-110 | 6 | 20 | N2 |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

**Pace Analytical Services, LLC**
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

## QUALIFIERS

Project: Biosolids
Pace Project No.: 50369587

### DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

### ANALYTE QUALIFIERS

| | |
|---|---|
| E | Analyte concentration exceeded the calibration range. The reported result is estimated. |
| F5 | The recovery of the analyte in the CRDL standard (also known as the reporting limit verification) did not meet the acceptance criteria. |
| H3 | Sample was received or analysis requested beyond the recognized method holding time. |
| L1 | Analyte recovery in the laboratory control sample (LCS) was above QC limits. Results for this analyte in associated samples may be biased high. |
| M0 | Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits. |
| M1 | Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery. |
| M3 | Matrix spike recovery was outside laboratory control limits due to matrix interferences. |
| N2 | The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request. |
| P1 | Routine initial sample volume or weight was not used for extraction, resulting in elevated reporting limits. |
| P6 | Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level. |
| R1 | RPD value was outside control limits. |
| S2 | Surrogate recovery outside laboratory control limits due to matrix interferences (confirmed by similar results from sample re-analysis). |

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

## QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Biosolids
Pace Project No.: 50369587

| Lab ID | Sample ID | QC Batch Method | QC Batch | Analytical Method | Analytical Batch |
|---|---|---|---|---|---|
| **50369587001** | **DIGESTOR #4** | EPA 3546 | 783413 | EPA 8081 | 783525 |
| **50369587001** | **DIGESTOR #4** | EPA 3510 | 784116 | EPA 8081 | 784261 |
| **50369587001** | **DIGESTOR #4** | EPA 3546 | 784646 | EPA 8082 | 784732 |
| **50369587001** | **DIGESTOR #4** | EPA 8151 | 784552 | EPA 8151A | 784743 |
| **50369587001** | **DIGESTOR #4** | EPA 3050 | 783830 | EPA 6010 | 784277 |
| **50369587001** | **DIGESTOR #4** | EPA 3010 | 784153 | EPA 6010 | 784212 |
| **50369587001** | **DIGESTOR #4** | EPA 3050B | 784173 | EPA 6020 | 784576 |
| **50369587001** | **DIGESTOR #4** | EPA 7470 | 784349 | EPA 7470 | 784413 |
| **50369587001** | **DIGESTOR #4** | EPA 7471 | 784384 | EPA 7471 | 784553 |
| **50369587001** | **DIGESTOR #4** | EPA 3546 | 783934 | EPA 8270 | 784139 |
| **50369587001** | **DIGESTOR #4** | EPA 3510 | 784240 | EPA 8270 | 784460 |
| **50369587001** | **DIGESTOR #4** | EPA 8260 | 783728 | | |
| **50369587001** | **DIGESTOR #4** | EPA 5030/8260 | 784084 | | |
| **50369587001** | **DIGESTOR #4** | SM 2540G | 783511 | | |
| **50369587001** | **DIGESTOR #4** | SM 2540G | 783512 | | |
| **50369587001** | **DIGESTOR #4** | SM 2540G | 783513 | | |
| **50369587001** | **DIGESTOR #4** | EPA 9038 | 783903 | EPA 9038 | 783911 |
| **50369587001** | **DIGESTOR #4** | EPA 9045 | 784239 | | |
| **50369587001** | **DIGESTOR #4** | EPA 351.2 | 783628 | EPA 351.2 | 783797 |
| **50369587001** | **DIGESTOR #4** | EPA 353.2 | 784583 | EPA 353.2 | 784584 |
| **50369587001** | **DIGESTOR #4** | SM 4500-NH3 B | 785030 | SM 4500-NH3 G | 785135 |
| **50369587001** | **DIGESTOR #4** | SM 4500-Cl-E | 784900 | SM 4500-Cl-E | 784938 |
| **50369587001** | **DIGESTOR #4** | SM 4500-CN-C | 784844 | SM 4500-CN-E | 785080 |

Pace® Location Requested (City/State):
Pace Analytical Grand Rapids
4171 40th Street SE, Grand Rapids, MI 49512

# CHAIN-OF-CUSTODY Analytical Request Document

Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevant fields

LAB USE ONLY- Affix Workorder/Login Label Here

WO#: 50369587

50369587

| | |
|---|---|
| Company Name: | City of Cadillac |
| Street Address: | 1121 Plett Rd, Cadillac, MI 49601 |
| Customer Project #: | |
| Project Name: | Biosolids |
| Site Collection Info/Facility ID (as applicable): | |
| Contact/Report To: | Tomaszewski, Cindy |
| Phone #: | 231-775-2368 |
| E-Mail: | ctomaszewski@cadillac-mi.net |
| Cc E-Mail: | |
| Invoice To: | SAME |
| Invoice E-Mail: | |
| Purchase Order # (if applicable): | |
| Quote #: | |
| County / State origin of sample(s): | Michigan |

Time Zone Collected: [ ] AK [ ] PT [ ] MT [ ] CT [ ] ET

Data Deliverables: [ ] Level II [ ] Level III [ ] Level IV [ ] EQUIS [ ] Other

Regulatory Program (DW, RCRA, etc.) as applicable: NPDES — Reportable [X] Yes [ ] No

Rush (Pre-approval required): [ ] Same Day [ ] 1 Day [ ] 2 Day [ ] 3 Day [ ] Other ____________

DW PWSID # or WW Permit # as applicable:

Date Results Requested:

Field Filtered (if applicable): [ ] Yes [X] No

Analysis:

* Matrix Codes (Insert in Matrix box below): Drinking Water (DW), Ground Water (GW), Waste Water (WW), Product (P), Soil/Solid (SS), Oil (OL), Wipe (WP), Tissue (TS), Bioassay (B), Vapor (V), Surface Water (SW),Sediment (SED), Sludge (SL), Caulk (CK), Leachate (LL), Biosolid (BS), Other (OT)

Specify Container Size **: 3, 1, 1, 4

Identify Container Preservative Type***: 1, 1, 1, 1

**Container Size: (1) 1L, (2) 500mL, (3) 250mL, (4) 125mL, (5) 100mL, (6) 40mL vial, (7) EnCore, (8) TerraCore, (9) 90mL, (10) Other

*** Preservative Types: (1) None, (2) HNO3, (3) H2SO4, (4) HCl, (5) NaOH, (6) Zn Acetate, (7) NaHSO4, (8) Sod. Thiosulfate, (9) Ascorbic Acid, (10) MeOH, (11) Other

Lab Use Only: Proj. Mgr: **Brian Hall**; AcctNum / Client ID: ; Table #: ; Profile / Template: **8311**; Prelog / Bottle Ord. ID: **1077504**

Analysis Requested

| Customer Sample ID | Matrix * | Comp / Grab | Composite Start Date | Composite Start Time | Collected or Composite End Date | Collected or Composite End Time | # Cont. | Res. Chlorine Results | Res. Chlorine Units | Metals/Wet Chem | SVOC,Pest/PCBs | TCLP V,SV,Pest,Herb,Met | VOC | Sample Comment | Preservation non-conformance identified for sample. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DIGESTOR#4 | SL | G | — | — | 4-1-24 | 1415 | 4 | — | | X | X | X | X | | |
| | | | | | | | | | | | | | | | |
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| | | | | | | | | | | | | | | | |

Additional Instructions from Pace®:
**Wet Chem: TKN,NH3,CN,Cl,NO2/NO3,SO4,pH,%Solid,TVS**

Collected By: (Printed Name) DAVID SKIFF

Signature: CATomaszewski for DS

Customer Remarks / Special Conditions / Possible Hazards:

| # Coolers: | Thermometer ID: | Correction Factor (°C): | Obs. Temp. (°C) | Corrected Temp. (°C) | On Ice: |
|---|---|---|---|---|---|
| | | | | | |

| Relinquished by/Company: (Signature) | Date/Time: | Received by/Company: (Signature) | Date/Time: |
|---|---|---|---|
| CATomaszewski | 4-1-2024/1500 | Ship via UPS Ground | |
| | | LynnRiegel/Pace | 4-2-24 10:04 |
| | | | |
| | | | |

Tracking Number:

Delivered by: [ ] In- Person [ ] Courier [ ] FedEX [ ] UPS [ ] Other

Page: of

Submitting a sample via this chain of custody constitutes acknowledgment and acceptance of the Pace® Terms and Conditions found at https://www.pacelabs.com/resource-library/resource/pace-terms-and-conditions/

ENV-FRM-CORQ-0019_v02_110123 ©

Pace

# Sample Conditions Upon Receipt Form (SCUR)

| | | | |
|---|---|---|---|
| Date/Time: 4/2 1226 | Evaluated By: DAS | **WO#: 50369587** PM: BJH Due Date: 04/16/24 CLIENT: GR-Cadillac | |
| Client: City of Cadillac | PM: BJH | | |
| Lab Notified of Rush or Short Holds: YES NO NA | | | |

| Project Received Via: FedEx UPS Client Pace Courier Other: ______ | | | | Comments: |
|---|---|---|---|---|
| Custody Seal Present and Intact: | YES | NO | NA | |
| Received Sample Information Form (SIF): Drinking Waters Only | YES | NO | NA | |
| Short Hold Present ( ≤ 48 Hours): | YES | NO | | |
| Sample Received in Hold: | YES | NO | | |
| Custody Signature Present: | YES | NO | | |
| Collector Signature Present: | YES | NO | | |
| Sample Collected Today and On Ice: | YES | NO | N/A | |
| IR Gun #: 350 351 Therm #: 282 283 | Temp. should be 0°C - 6°C (Initial/Corrected) | | | |
| Ice Type: WET Bagged / WET Loose BLUE NONE | 1. Cooler Temp. Upon Receipt: 4.2/3.8 °C | | | |
| Ice Location: TOP BOTTOM MIDDLE DISPERSED | 2. Cooler Temp. Upon Receipt: ______ °C | | | |
| Temp Blank Received: | YES | NO | | |
| Sample Label Matches COC (ID/Date/Time): | YES | NO | | |
| Container Intact: | YES | NO | | |
| Correct Container: | YES | NO | | |
| Sufficient Volume: | YES | NO | | |
| Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with red dot on container lid if unacceptable. pH Strip Lot #: ______ *Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl* | YES | NO | N/A | |
| Residual Chlorine Absent: $Cl_2$ Strip Lot #: ______ *Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide* | YES | NO | N/A | |
| VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable. | YES | NO | N/A | |
| Trip Blank Received: HCl MeOH Other: ______ | YES | NO | ON HOLD | |
| Comments: | 3. Cooler Temp. Upon Receipt: ______ °C | | | |
| | 4. Cooler Temp. Upon Receipt: ______ °C | | | |
| | Non-Conformance Form Required: YES NO | | | |